Quartic-width learning bounds for robust Bayesian tanh networks

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Abstract

We prove sample bounds for Gaussian Bayesian tanh networks restricted to predictions stable under simultaneous internal disturbances. For a layered Boolean teacher of width C and $L-1$ gate stages, a sufficient sample count is $\widetilde{O}(C^4L^{11})$, uniformly over every learner width $N \geq C$ and every mean-field-or-stronger scale $\gamma \geq 1$. For $\gamma \geq 8/7$ the bound improves to $\widetilde{O}(C^4L^9)$. A separate stronger-scale argument gives $\widetilde{O}(\min\{C^4L^7, C^{9/2}L^6\})$ for $\gamma \geq 2$ and in the strong-prior limit at each fixed width. The robustness restriction is the same in all these statements. Input dimension, robustness calibration, accuracy, and confidence are fixed in these rate displays. The all-scale proof uses an entropy-weighted regression and a positive estimate of the Gaussian variance to construct a smaller finite comparison class. A direct Bayesian bound handles widths where the information needed to describe that construction is too costly. The stronger-scale proof uses multilevel sampling of two consecutive layers. We give the model, exact budgets, new proofs, and explicitly state the previously established reconstruction estimates used. These are conventional mathematical proofs, not Lean formalizations.

1 Introduction

A polynomial sample bound can still be very large. For a neural network learning a Boolean circuit, it matters how the bound depends jointly on the circuit's width and depth. We study exact Gaussian Bayesian prediction with tanh hidden layers and a linear readout, conditioned on noiseless training margins. The prior retains only networks whose clipped outputs are stable under a calibrated ball of simultaneous internal disturbances. The restriction is stated in full below and is fixed before observing training labels.

The main result is quartic dependence on teacher width throughout the mean-field-or-stronger Gaussian regime: $\widetilde{O}(C^4L^{11})$ samples suffice at every learner width $N \geq C$. At moderately stronger scales the bound is $\widetilde{O}(C^4L^9)$. At still stronger scales, a different method gives a choice between $\widetilde{O}(C^4L^7)$ and $\widetilde{O}(C^{9/2}L^6)$. The latter improves depth dependence at a small cost in the width exponent; it is not a quartic-width depth-six bound. All statements allow an arbitrary common iid training and test input distribution.

The all-scale construction measures approximation error using a matrix that records the conditional Gaussian entropy in different active directions. Independent-batch regression controls error in that metric. An upper estimate of the total variance then permits a stable nonlinear projection and just one Gaussian coefficient per reconstructed row. The resulting comparison functions are finite arithmetic programs; they need not be neural networks. Only the original posterior networks must

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satisfy the stated robustness restriction. A direct prior-mass bound handles the remaining widths and also sharpens the dependence on the Gaussian scale.

The second construction approximates two consecutive activation layers using antithetic multilevel sampling. Its accuracy and cost depend on aggregate squared weights. Allocating spectral cutoffs across layers using their individual energies yields the depth-six tradeoff. Neither construction adds a merging, coherence, or spectral-product condition to the prior.

After the model and main statements, we give the simpler energy-based code and its posterior argument. The appendices provide the target cylinder, the prior reconstruction and density estimates, and the three new arguments: entropy-metric compression (Appendix D), the direct posterior and scale interpolation bound (Appendix E), and multilevel compression (Appendix F). The supporting energy and factor codes are retained to make their exact hypotheses and their role in the new bounds explicit; those supporting codes build on [1]. No efficient posterior inference algorithm or optimality of the sample powers is asserted.

2 Learner, target, and robustness restriction

Let $d \geq 1$, $L \geq 2$, and $H = L - 1$. Inputs lie in $\mathcal{X} = \{-1, 1\}^d$, with arbitrary distribution D. A learner of hidden width N has parameters $w = (W^1, \ldots, W^H, v)$ and computes

$$h^1 = \tanh(W^1 \bar{x}), \qquad h^s = \tanh(W^s h^{s-1}), \qquad z_w(x) = v^\top h^H, \qquad \bar{x} = (x, 1).$$

The raw law $\mu_{N,t}$ has independent centered Gaussian entries of variance t^2/d in W^1, t^2/N in the internal matrices, and t^2/N^2 in v. Here $0 < t \leq 1$. The scale may be any prescribed function $t = t(d, L, N)$; it need not be constant as width varies. In particular $t(N) = N^{(1-\gamma)/2}$ gives variances $N^{1-\gamma}/d$, $N^{-\gamma}$, and $N^{-\gamma-1}$.

Define weighted energy, readout energy, and clipped score by

$$\kappa_N(w) = \frac{d}{N} \left\| W^1 \right\|_F^2 + \sum_{s=2}^{H} \|W^s\|_F^2 + N \|v\|_2^2 ,$$
$$E_v(w) = N \|v\|_2^2 , \qquad T(u) = \max\{-1/2, \min\{1/2, u\}\}.$$

Disturbances are added after each activation and at the readout:

$$h^1_\eta = \tanh(W^1 \bar{x}) + \eta^1, \quad h^s_\eta = \tanh(W^s h^{s-1}_\eta) + \eta^s, \quad z_w(x; \eta) = v^\top h^H_\eta + \eta^{\text{out}}.$$

Their norm is

$$\|\eta\|_*^2 = \sum_{s=1}^{H} \frac{\|\eta^s\|_2^2}{N} + |\eta^{\text{out}}|^2.$$

Fix $c > 0$ before seeing labels. The support condition is

$$\mathcal{R}_{N,c} = \left\{ w : |T(z_w(x; \eta)) - T(z_w(x))| \leq \frac{1}{16} \text{ for all } x \in \mathcal{X}, \quad \|\eta\|_* \leq \frac{c}{\sqrt{1 + H E_v(w)}} \right\}. \tag{1}$$

Noisy hidden values are allowed outside $[-1, 1]^N$. This is a label-independent Borel condition, since the input set is finite and the response is continuous in weights and disturbances.

For a sample $S = ((X_i, f(X_i)))_{i=1}^n$ with iid $X_i \sim D$, let $Q_{S,N,t,c}$ be the raw Gaussian conditioned on

$$w \in \mathcal{R}_{N,c}, \qquad f(X_i) z_w(X_i) > 1/2 \quad (1 \leq i \leq n). \tag{2}$$

There is no initial clean-output-margin condition on unobserved inputs. The risk is posterior-average test-margin failure,

$$R(Q) = \mathbb{E}_{w\sim Q}\mathbb{P}_{X\sim D}\{f(X)z_w(X) \le 1/4\}.$$

The target is a single-output Boolean circuit of width $C \ge 2$ and H layered stages, using NAND, COPY, and NOT. A constant-one wire is reserved at every stage and counted toward C; compilation overhead from another basis is also counted. Unused slots and stages may contain constants or copies. Thus a teacher with $h \le H$ stages can be padded to the learner's depth. If the learner matches the teacher's depth, $L = h + 1$; if it is deeper, its actual L occurs in the bound. Width independence does not mean depth independence.

3 Learning results

Theorem 1 (Quartic width throughout the good Gaussian regimes)**.** *Fix $d, c, \varepsilon, \delta$ with $c > 0$ and $0 < \varepsilon, \delta < 1$. Put $F_0 = 1 + \log^2(2 + C)$ and $G = \log(2 + CL)$. There is a constant A_0, depending only on these four quantities, such that*

$$n \ge A_0 C^4 L^{11} F_0^4 G^2$$

implies, with sample probability at least $1 - \delta$, $R(Q_{S,N,t,c}) \le \varepsilon$ simultaneously for every $N \ge C$ and every $0 < t \le 1$.

More precisely, for each $\gamma \ge 1$ it suffices to require

$$n \ge A_0 F_0^4 G^2 \left[C^4 L^9 + C^{(6\gamma-2)/(3\gamma-2)} L^{(13\gamma-2)/(3\gamma-2)}\right]$$

to obtain the same conclusion simultaneously for $N \ge C$ and $0 < t^2 \le N^{1-\gamma}$. In particular, $A_0 C^4 L^9 F_0^4 G^2$ samples suffice for $\gamma \ge 8/7$. The constant can be chosen independently of γ.

The statements hold for every Boolean teacher and iid input law specified above. They also hold after adding the label-independent prior restriction $|z_w(x)| > 1/2$ at every Boolean input.

Proof and exact-budget specification. Use the product target cost in (17) and the posterior budgets in (18). The readout-tail calculation in Appendix C retains posterior mass at least $1 - \beta$ on a common robustness radius r. Its retained law Q' satisfies $Q' \le e^{NK/t^2}\mu_{N,t}$. Apply Theorem 14 to this law. The ramp comparison in Appendix C gives precisely (41), with the same α, β. For the complementary information range use the raw posterior density bound and Lemma 16. Theorem 17 then applies with the finite code (EM25) and threshold $\Delta = \max\{K, 2LD_0\}$.

At fixed d, c, ε, the explicit budgets give

$$K = O(CLF_0), \quad r^{-2} = O(CL^2F_0), \quad e^{-1} = O(CL^3F_0), \quad \eta^{-1} = O(L),$$

$$q = O(C^2L^4F_0^2), \qquad m = O([C^2L^4F_0^2 + t_0^4C^2L^6F_0^2]G).$$

The scalar grid costs $O(G)$ per coefficient, and the entropy-matrix message costs $O(q^2G)$ per layer. Thus the constants in (40) may be chosen with

$$A = O(C^4L^9F_0^4G^2), \qquad B = O(C^4L^{11}F_0^4G^2).$$

The remaining finite-family confidence charge is logarithmic. Equations (45) and (46) prove the two claims. The exponents of F_0 and G in the interpolated term are at most four and two, respectively, uniformly in $\gamma \ge 1$. Its width exponent is at most four, and its depth exponent is at most nine when $\gamma \ge 8/7$. The same target cylinder survives optional global-margin conditioning, so the identical denominator, tail, and coupling arguments apply. □

Theorem 2 (A further depth improvement at stronger scales)**.** *At fixed $d, c, \varepsilon, \delta$, the same robust Bayesian learner has sufficient sample order*

$$\widetilde{O}\Big(\min\{C^4L^7, C^{9/2}L^6\}\Big)$$

simultaneously for every $N \geq C$ and $0 < t \leq N^{-1/2}$, including every $\gamma \geq 2$. The same order suffices in the strong-prior limit at each fixed width. Both conclusions refer to posterior-average test-margin failure $R(Q) \leq \varepsilon$ with sample probability at least $1 - \delta$.

The full finite-scale alternatives, sample budget, and proof are in Theorem 21. The depth-six branch improves the powers when $C < L^2$. The baseline quartic code and its probability calculation are given next; the multilevel construction replaces only that code.

3.1 The energy-based baseline and explicit stronger scales

The following separate argument trades its stronger scale hypothesis for quartic dependence on teacher width. Define

$$\begin{aligned}
e_* &= \frac{1}{16\sqrt{2C}}, \qquad \tau_0 = e_*/4, \\
b_L &= \sqrt{16\log 9 + 8\log(4L)}, \\
A &= \max\{32b_L, 2\log(8/e_*)\}, \qquad a = \max\{2, 256c\}, \\
\rho &= \min\{1/4, A/[4(d+1)], Ae_*/(64\sqrt{C})\}, \\
q_0 &= d + 2 + (L-2)C, \\
B_0 &= 9dA^2 + 18(L-2)CA^2 + 2a^2C, \\
D_* &= B_0 + q_0 \log\frac{e\sqrt{2\pi}}{2\rho} + \log 2.
\end{aligned} \tag{3}$$

These constants provide a target-realizing subset of the prior support of raw mass at least $\exp(-ND_*/t^2)$ at every $N \geq C$; the proof is in Appendix A. In particular $D_* \geq 1$ and every posterior in the theorem below is well-defined.

For $K \geq 1$, set

$$\begin{aligned}
r &= \min\{1/6, c/\sqrt{1+HK}\}, & m &= \left\lceil 256Kr^{-2}[d + \log(400L)] \right\rceil, \\
B &= \sqrt{4mK}, & \zeta &= \frac{r}{4\sqrt{(d+1)^2 + Hm^2}}, \\
P &= m(d+1) + (L-2)m^2 + m, & J &= P\log(3 + 2B/\zeta).
\end{aligned} \tag{4}$$

Theorem 3 (Uniform-width learning with polynomial coding precision)**.** *Let $0 < \varepsilon, \delta < 1$ and suppose f has the Boolean implementation specified above. The sample condition*

$$n \geq \frac{2}{\varepsilon}\left[J + \log(1/\delta)\right] \tag{5}$$

suffices in either of the following cases:

1. *Set $K = 8D_*$. Allow every $N \geq C$ and every positive scale satisfying*

$$t^2 \leq \min\left\{1, \frac{ND_*}{(P_N/2)\log 2 + \log(2/\varepsilon)}\right\}, \qquad P_N = N(d+1) + (L-2)N^2 + N.$$

2. *Set* $K = 8D_* + 4(d+L) + 4\log(2/\varepsilon)$. *Allow every* $N \geq C$ *and* $0 < t \leq N^{-1/2}$, *including every fixed exponent* $\gamma \geq 2$.

With probability at least $1-\delta$ *over the sample, the risk* $R(Q_{S,N,t,c})$ *is at most* ε *simultaneously for all indicated widths and scales.*

For every fixed width, the first clause permits all sufficiently small positive scales and therefore gives the strong-prior risk conclusion $\limsup_{t\downarrow 0} R(Q_{S,N,t,c}) \leq \varepsilon$ on the same sample event. No identification of a separately defined zero-temperature posterior is needed. If σ denotes the strong-prior parameter in the convention with density proportional to $\exp(-\kappa_N/(2\sigma^2))$, then $t = \sqrt{N}\,\sigma$.

Corollary 4 (Explicit width and depth powers)**.** *For fixed* $d, c, \varepsilon, \delta$, *put*

$$F = 1 + \log^2(2+C) + \log(2+L), \qquad G = \log(2+CL).$$

In either scale regime of Theorem 3, a sufficient sample count has order

$$O\Big(C^4 L^7 F^4 \log^2(2+L) G\Big) = \widetilde{O}(C^4 L^7). \tag{6}$$

At fixed depth this is $O(C^4 \log^9(2+C))$; *at fixed width it is* $O(L^7 \log^7(2+L))$.

The exact statistical dependence is in (4)–(5). In the second scale regime accuracy also enters K through $\log(2/\varepsilon)$. The asymptotic display (6) holds other resources fixed; it is not an optimality claim.

For the usual confidence criterion, define $p_S(x) = Q_{S,N,t,c}\{f(x)z_w(x) > 1/4\}$. Then $D\{p_S \leq 2/3\} \leq 3R(Q)$; using $\varepsilon/3$ in the theorem gives correct-margin posterior probability strictly greater than $2/3$ on all but an ε fraction of test inputs. This also gives correct-sign confidence and covers arbitrary posterior mixtures.

4 A quantized sampling construction

We prove the deterministic finite-code statement needed for the posterior. Fix N and a network w with $\kappa_N(w) \leq K$. Write $k_s = \|W^s\|_F^2$ for $2 \leq s \leq H$ and $k_L = N\,\|v\|_2^2$. Their sum is at most K.

4.1 Pruning and conditional concentration

In every internal row and in the readout, remove entries with absolute value greater than

$$\frac{8\sqrt{K}\;\|\mathrm{row}\|_2}{r\sqrt{N}}.$$

Zero rows remain zero. Denote pruned arrays by $\bar{W}^s$ and $\bar{v}$. The absolute mass removed from a row is at most its squared norm divided by this threshold. On any vector in $[-1,1]^N$, the internal pruning error therefore has Euclidean norm at most $r\sqrt{Nk_s/K}/8$; the readout error is at most $r\sqrt{k_L/K}/8$.

Let $S^1, \ldots, S^H$ be independent lists of m uniform indices in $\{1, \ldots, N\}$, with replacement. Let $\mathcal{Q}_\zeta$ round a real number to a nearest multiple of ζ, with a fixed tie convention. It preserves zero

and has error at most ζ. Define the *full virtual states* by

$$\begin{aligned}
\widehat{h}_i^1(x) &= \tanh\left(\sum_{j=1}^{d+1} \mathcal{Q}_\zeta(W_{ij}^1)\bar{x}_j\right), \\
\widehat{h}_i^s(x) &= \tanh\left(\sum_{j=1}^{m} \mathcal{Q}_\zeta\left(\frac{N}{m}\bar{W}^s_{i,S_j^{s-1}}\right)\widehat{h}^{s-1}_{S_j^{s-1}}(x)\right), \quad 2 \le s \le H, \\
\widehat{z}(x) &= \sum_{j=1}^{m} \mathcal{Q}_\zeta\left(\frac{N}{m}\bar{v}_{S_j^H}\right)\widehat{h}^H_{S_j^H}(x).
\end{aligned} \tag{7}$$

The stored coordinates at the lists S^s form an ordinary width-m tanh network. This holds even when $m > N$ or indices repeat.

For each $s \ge 2$, the state $\widehat{h}^{s-1}$ depends only on $S^1, \dots, S^{s-2}$. It remains in $[-1,1]^N$ and is independent of the fresh list S^{s-1}. Conditional on the preceding lists, for one uniform index I define

$$Y = N\bar{W}^s_{:,I}\widehat{h}^{s-1}_I.$$

The pruning threshold and bounded states give

$$\mathbb{E}\,\|Y - \mathbb{E}Y\|_2^2 \le Nk_s, \qquad \|Y - \mathbb{E}Y\|_2 \le 16\sqrt{NKk_s}/r.$$

Hilbert-space Bernstein [5], applied at tolerance $r\sqrt{Nk_s/K}/8$, bounds the failure probability by

$$2\exp\left(-\frac{3mr^2}{640K}\right) \le 2\exp\left(-\frac{mr^2}{256K}\right).$$

The readout uses the independent final list S^H and has the same calculation without factors $\sqrt{N}$. Zero-energy stages are exact. A union bound over the 2^d inputs and at most L stages gives simultaneous success probability at least 199/200 by (4). The concentration estimate is applied *before* coefficient rounding at the current stage, but at the already rounded preceding virtual state. This is why its conditional independence remains valid.

4.2 All errors are disturbances of the original network

On this concentration event, pruning and sampling together give stage preactivation errors at most $r\sqrt{k_s/K}/4$ in normalized Euclidean norm, with the scalar counterpart at the readout. Tanh is 1-Lipschitz, so the corresponding stacked postactivation/readout error has norm at most $r/4$.

Rounding adds at most $(d+1)\zeta$ per first-layer coordinate, and at most $m\zeta$ per coordinate in every later layer and at the readout. This uses only that all virtual hidden states in (7) lie in $[-1,1]^N$. The stacked rounding error thus satisfies

$$\|\eta_{\text{round}}\|_*^2 \le \zeta^2\left[(d+1)^2 + Hm^2\right] = r^2/16.$$

Define the actual disturbances in the original, unpruned network by

$$\eta^1 = \widehat{h}^1 - \tanh(W^1\bar{x}), \qquad \eta^s = \widehat{h}^s - \tanh(W^s\widehat{h}^{s-1}), \qquad \eta^{\text{out}} = \widehat{z} - v^\top\widehat{h}^H.$$

They reproduce the virtual states and output exactly. Splitting each into the pruning/sampling and rounding contributions and using the triangle inequality in the stacked norm gives

$$\|\eta\|_* \le r/4 + r/4 = r/2. \tag{8}$$

If $w \in \mathcal{R}_{N,c}$, then $r \le r_c(w)$ because $E_v(w) \le K$. The original robustness assumption therefore yields

$$\sup_{x \in \mathcal{X}} |T(z_w(x)) - T(\widehat{z}(x))| \le 1/16. \tag{9}$$

There is no subsequent unprotected rounding step, and no assertion that the sampled network itself satisfies (1).

4.3 A finite code independent of learner width

Consider the same sampled coefficients before applying $\mathcal{Q}_\zeta$. Their energy expectations obey

$$\frac{d}{m}\mathbb{E}\left\|W^1_{\mathrm{comp}}\right\|_F^2 = \frac{d}{N}\left\|W^1\right\|_F^2, \qquad \mathbb{E}\left\|W^s_{\mathrm{comp}}\right\|_F^2 \le \|W^s\|_F^2, \qquad m\mathbb{E}\left\|v_{\mathrm{comp}}\right\|_2^2 \le N\|v\|_2^2.$$

These statements concern sampled parameters, not virtual activations. They use independence of adjacent index lists and are unaffected by rounding inside (7). Markov's inequality gives sampled unrounded energy at most $4K$ with probability at least $3/4$. This event intersects the concentration event.

On the intersection, every unrounded stored coefficient has absolute value at most $B = \sqrt{4mK}$. Its rounded value belongs to a fixed grid containing at most $3 + 2B/\zeta$ points. Although rounding was defined on all real numbers for the full virtual computation, only these bounded stored coefficients enter its width-m realization. Thus a fixed class $\mathcal{G}$ of rounded networks satisfies

$$\log|\mathcal{G}| \le P \log(3 + 2B/\zeta) = J, \tag{10}$$

and every internally robust energy-K network at every width has a member of this class satisfying (9).

5 Posterior learning and the sample polynomial

Proof of Theorem 3. Let ψ equal one on $(-\infty, 1/3]$, zero on $[5/12, \infty)$, and be affine between these thresholds. Every robust network of energy at most K has a rounded representative $g \in \mathcal{G}$. If it fits a training margin, then

$$f(X_i)T(g(X_i)) \ge 1/2 - 1/16 = 7/16 > 5/12.$$

Every test input where $f(x)z_w(x) \le 1/4$ satisfies $f(x)T(g(x)) \le 1/4 + 1/16 = 5/16 < 1/3$. The representative therefore has zero empirical ramp loss, and its population ramp loss bounds the original network's test-margin error.

For a fixed nonnegative loss bounded by one with population mean u, the chance that all n iid losses equal zero is at most $(1-u)^n \le e^{-nu}$. A union bound using (10) and (5) shows that, with probability at least $1 - \delta$, every zero-empirical-loss representative has population loss at most $\varepsilon/2$. This single sample event controls every robust fitting network of energy at most K, uniformly over width and scale.

The target cylinder bounds the raw conditioning probability below by e^{-ND_*/t^2} for every correctly labeled sample. Under the raw Gaussian, $N\kappa_N/t^2$ is chi-square with P_N degrees of freedom. Its exponential moment gives

$$Q_{S,N,t,c}\{\kappa_N > K\} \le \exp\left(-\frac{NK}{4t^2} + \frac{P_N}{2}\log 2 + \frac{ND_*}{t^2}\right). \tag{11}$$

For $K = 8D_*$, the first scale threshold makes this at most $\varepsilon/2$. For the second choice of K, use $t^2 P_N/N \le d + L$ when $t^2 \le 1/N$. The exponent multiplied by t^2/N is at most

$$-K/4 + D_* + (d+L)(\log 2)/2 \le -\log(2/\varepsilon).$$

Since $N/t^2 \ge 1$, the tail is again at most $\varepsilon/2$. Adding this discarded posterior mass to the retained risk proves the claim for arbitrary posterior mixtures. □

Proof of Corollary 4. Put $T_L = d + \log(400L)$. The budget gives $m = O(Kr^{-2}T_L)$. All factors in B/ζ are polynomial in d, L, K, r^{-1}, so

$$J = O\Big([dKr^{-2}T_L + LK^2 r^{-4} T_L^2] \log(2 + d + L + K + r^{-1})\Big).$$

The target constants imply

$$D_* = O\Big((d + LC)[\log^2(2C) + \log(2L) + \log(2 + d + C)] + (1 + c^2)C\Big).$$

At fixed $d, c, \varepsilon, \delta$, it follows that

$$K = O(CLF), \quad r^{-2} = O(CL^2F), \quad m = O(C^2L^3F^2 \log(2+L)),$$

$$P = O(C^4L^7F^4 \log^2(2+L)), \qquad \log(3 + 2B/\zeta) = O(G).$$

Substitution proves (6) and its fixed-width and fixed-depth consequences. □

6 Scope of the two compression arguments

The energy-based baseline just proved has a total-energy cutoff. At mean-field scaling, typical internal Frobenius energy grows with learner width, so that particular proof cannot give its stated quartic rate throughout the scale range. Entropy-metric compression avoids a width-independent total-energy cutoff and proves the all-scale theorem instead. Both methods use the same prior support.

The new depth-six tradeoff and all-scale quartic bound are sufficient rates. Quadratic teacher-width dependence under this unchanged support, and a depth-linear bound with quartic width, remain open. No matching lower bound for the displayed polynomial exponents is asserted. The proofs build on the target-cylinder and reconstruction estimates of [3, 2, 4]; their nontrivial imported lemmas are stated explicitly below. These new arguments and rates are not Lean formalized. Existing formal proofs for earlier bounds do not certify them.

A A robust target cylinder at every width

We give the probability bound used in the statistical proof. It uses a robust digital implementation, not a claim that every implementation is stable. Let $N \ge C$. Partition the neurons into C balanced clone groups of sizes $k_a \in \{\lfloor N/C \rfloor, \lceil N/C \rceil\}$, so $C/2 \le N/k_a \le 2C$. Let P project onto vectors constant on each clone group. Fill unused logical slots with available constants or copies of an existing Boolean wire. The ideal wire vector $b^s(x)$ then has entries in $\{-1, 1\}$, belongs to the range of P, and has Euclidean norm $\sqrt{N}$.

Use gain A in the tanh simulation of every Boolean gate. For example, NAND on $u, v \in \{-1, 1\}$ is the sign of $1 - u - v$; copies and negations use u and $-u$. Bias constants in internal layers are supplied by counted constant wires. The ideal signed preactivation of every gate is at least A. In

the clone implementation, each coefficient from a logical wire is divided equally among that wire's clones. Set

$$e_* = \frac{1}{16\sqrt{2C}}, \qquad \tau_0 = \frac{e_*}{4}, \qquad b_L = \sqrt{16\log 9 + 8\log(4L)},$$
$$A = \max\{32b_L, 2\log(8/e_*)\}, \quad a = \max\{2, 256c\}, \quad \rho = \min\{1/4, A/[4(d+1)], Ae_*/(64\sqrt{C})\}.$$

The teacher readout averages the clones of the output wire with total coefficient a.

In Gaussian coordinates $\sqrt{d}\,W^1$, $\sqrt{N}\,W^s$, and Nv, all entries have variance t^2. Restrict each first-layer coordinate, each internal row's projection onto every normalized clone-group direction, and each readout coordinate to an interval of radius ρt about its teacher value. The internal matrices then decompose as

$$W^s = W^s_* + R^s + E^s, \qquad R^s = R^s P, \quad \|R^s\|_{\mathrm{op}} \le \rho t\sqrt{C}, \quad E^s P = 0.$$

The remaining Gaussian bulk E^s is independent of the constrained coordinates. A 1/4-net of the unit sphere has at most 9^N points, and the usual two-net Gaussian bound gives

$$\Pr\{\|E^s\|_{\mathrm{op}} > b_L t\} \le 2\exp(2N\log 9 - Nb_L^2/8) \le \frac{1}{2L}.$$

Thus all bulk norms are at most $b_L t$ with conditional probability at least 1/2.

Finite disturbance robustness. Allow a separate RMS disturbance of size τ_0 at each hidden layer and scalar output disturbance at most τ_0. This includes the entire total-budget ball $\|\eta\|_* \le \tau_0$. Write $e_s = \|h^s_\eta - b^s\|_2/\sqrt{N}$. If $e_{s-1} \le e_*$, every clone-group mean differs from its ideal bit by at most $e_*\sqrt{N/k_a} \le 1/16$. At most three teacher coefficients, including a constant coefficient, enter a gate, so its intended signed preactivation remains at least $A/2$. For $\epsilon \in \{-1, 1\}$ and real u, the elementary global secant bound

$$\epsilon q \ge A/2 \quad \Longrightarrow \quad |\tanh(q+u) - \epsilon| \le 2e^{-A/2} + 8|u|/A$$

holds: for $|u| < A/4$ use saturation, and otherwise the linear term is at least two. Since $E^s b^{s-1} = 0$ and $\|h^{s-1}_\eta\|_2/\sqrt{N} \le 1 + \tau_0$, it follows that

$$e_s \le 2e^{-A/2} + (8/A)\rho\sqrt{C}(1+\tau_0) + (8b_L/A)e_{s-1} + \tau_0 \le e_*.$$

Each of the four terms is at most $e_*/4$. The first layer initializes the same invariant: its coordinatewise preactivation perturbation is at most $\rho(d+1) \le A/4$, and its postactivation disturbance contributes at most τ_0 in RMS norm. Consequently the noisy signed readout is at least

$$f(x)z_w(x;\eta) \ge \frac{15a}{16} - \rho(1+\tau_0) - \tau_0 > 1 \qquad (x \in \mathcal{X},\ \|\eta\|_* \le \tau_0).$$

The readout center satisfies $\sqrt{N}\|v_*\| \ge a\sqrt{C/2}$, and its permitted perturbation has $\sqrt{N}\|v - v_*\| \le \rho$. Therefore $E_v \ge a^2 C/8$, whence

$$r_c(w) \le \frac{2\sqrt{2}c}{a\sqrt{C}} \le \frac{1}{64\sqrt{2C}} = \tau_0.$$

All clean and noisy clipped scores on the calibrated ball equal $f(x)/2$. The entire cylinder, with the bulk-norm event imposed, therefore belongs to $\mathcal{R}_{N,c}$ and satisfies the training likelihood for every sample. This also proves why every fixed $c > 0$ is allowed: a larger calibration requires only a polynomially larger readout coefficient.

Gaussian mass. There are Nq constrained independent scalar coordinates, where $q = d + 2 + (L-2)C$. The sum of their squared centers is at most NB_0, with

$$B_0 = 9dA^2 + 18(L-2)CA^2 + 2a^2C.$$

These loose constants cover the fan-in-two gate implementation above. For $Z \sim \mathcal{N}(0, t^2)$, $\rho \le 1/4$, and any center m,

$$\Pr\{|Z - m| \le \rho t\} \ge \frac{2\rho}{e\sqrt{2\pi}} \exp(-m^2/t^2).$$

Multiplication over the coordinates and the independent bulk event gives

$$\mu_{N,t}(\text{target cylinder}) \ge \exp(-ND_*/t^2), \qquad D_* = B_0 + q \log \frac{e\sqrt{2\pi}}{2\rho} + \log 2. \tag{12}$$

Here $D_* \ge 1$ and is bounded by a universal polynomial in d, L, C, c. The construction and estimate hold at every integer width $N \ge C$.

B Sequential quantization at every mean-field-or-stronger scale

The energy-based baseline compressor requires a bound on total network energy. At the finite mean-field scale, the unconditioned Gaussian background can have growing total energy. An entropy-based compressor avoids that difficulty. In this appendix we combine the same sequential-quantization principle with the active-direction and covariance reconstruction estimates of [4]. This supporting code gives a sufficient bound at every $0 < t \le 1$, including $\gamma = 1$; Appendix D gives the sharper headline bound. This appendix is not a new proof of those active-direction estimates: we state precisely which properties are imported and prove the modification that removes the coding precision's extra depth factor. Neither this modification nor its new depth exponent is Lean formalized.

B.1 Budgets and learning statement

Use the model, support $\mathcal{R}_{N,c}$, and posterior $Q_{S,N,t,c}$ from (1)–(2). Let D_* be any target-cylinder cost satisfying the conclusion of Appendix A. Thus the raw posterior conditioning event has probability at least $\exp(-ND_*/t^2)$, for every correctly labeled sample and every $N \ge C$, $0 < t \le 1$. For $0 < \varepsilon, \delta < 1$, put

$$\beta = \varepsilon/4, \qquad B_v = \max\{8, 8[D_* + \log(1/\beta)]\}, \qquad K = D_* + 1,$$
$$r = \min\{1/6, c/\sqrt{1 + (L-1)B_v}\}, \qquad \alpha = \varepsilon/12.$$

Define the compression parameters

$$e = \frac{\alpha r^2}{1000L}, \qquad \eta = \frac{\alpha}{100L}, \qquad q_a = \left\lceil \frac{2K}{e\eta} \right\rceil,$$
$$D_0 = [(d+1)q_a + 2]\log 2, \qquad V = 8K + 2q_a, \qquad B_H^2 = 4V/e,$$
$$K_0 = (4K + 2\log 2)/\eta, \qquad B_1^2 = 4[4K + 2(d+1)\log 2]/e.$$

The letter q_a denotes the number of retained active directions; it is unrelated to a population partition or a support restriction. Let m be the least integer no smaller than each of

$$10000(V + q_a + 1)/e,$$
$$64(q_a + 1)\log(4(q_a + 1)/\eta),$$
$$512K_0 r^{-2}[d\log 2 + \log(2/\eta) + 1].$$

Finally set

$$M = \max\{m, \lceil LD_0/K \rceil, 1\}, \qquad B_G = \max\{2, 2\sqrt{\log(6m/e)}\},$$

$$P_* = M(d+1) + (L-2)M^2 + M, \qquad B_{\text{small}}^2 = \frac{4(4MK + 2P_* \log 2)}{\alpha},$$

$$B_{\text{large}} = \max\left\{B_1, \frac{2B_H\sqrt{q_a+1}}{m} + \frac{\sqrt{2}B_G}{\sqrt{m}}, \frac{\sqrt{K_0}}{m}, 1\right\},$$

$$B_* = 1 + \max\{B_{\text{small}}, B_{\text{large}}, 1\}, \qquad \zeta = \frac{\sqrt{e}}{4(M+d+1)},$$

$$J_* = P_* \log(3 + 4B_*/\zeta).$$

All square roots above denote the nonnegative root.

Theorem 5 (An improved all-scale budget)**.** *Under the Boolean-teacher convention of the main theorem, the condition*

$$n \geq \frac{16}{\varepsilon}\left[J_* + \log(1/\delta)\right] \tag{13}$$

implies, with probability at least $1-\delta$ *over iid training inputs,*

$$\mathbb{E}_{w \sim Q_{S,N,t,c},\, X \sim D} \mathbf{1}\{f(X) z_w(X) \leq 1/4\} \leq \varepsilon$$

simultaneously for every $N \geq C$ *and every* $0 < t \leq 1$*. The conclusion is also valid if global clean-output-margin conditioning is added to the prior. No population-merging or additional robustness assumption is made.*

With $d, c, \varepsilon, \delta$ fixed, write

$$F = 1 + \log^2(2+C) + \log(2+L), \qquad G = \log(2+CL).$$

The target constants and the budgets above give

$$n = O(C^6 L^{17} F^6 G). \tag{14}$$

This improves the former $O(C^6 L^{18} F^6 G)$ budget while preserving its fixed-depth $O(C^6 \log^{13}(2+C))$ width consequence. A further target-cylinder sharpening below removes the $\log(2+L)$ term from F.

B.2 Imported entropy and reconstruction estimates

We use the following properties of the construction in the appendix "Finite posterior compression by internal-noise coupling" of [4]. Its lemmas on future-weight entropy, whole-layer active directions, and whitened sampling prove these properties for *any bounded virtual prefix*. They do not require a Gaussian law for the prefix and do not assume its robustness.

First, suppose a prefix is A, the current original weight block is X, and the future blocks are Y, with product Gaussian reference law $P_X \otimes P_Y$. For a finite message J, a retained quantity $H_0 = f(A, J, X)$, and a new prefix A' which is a randomized function of (A, J, H_0) using fresh randomness conditionally independent of Y,

$$\mathbb{E}_{A'} D_{\text{KL}}(Q_{Y|A'} \| P_Y) \leq \mathbb{E}_A D_{\text{KL}}(Q_{X,Y|A} \| P_X \otimes P_Y) + \mathcal{H}(J \mid A). \tag{15}$$

Here $\mathcal{H}$ denotes conditional Shannon entropy. The inequality is the KL chain rule and data processing; deterministic processing of A' cannot increase its left side.

Second, if $Q \ll \mu_{N,t}$ has $D_{\mathrm{KL}}(Q\|\mu_{N,t}) \le NK/t^2$ and $N > M$, the budgets above permit the following sequential construction. Its current virtual layer has N rows, each evaluated on m stored source neurons from the preceding layer. The first original layer is clipped rowwise at physical standardized norm B_1. Each internal stage retains at most q_a active input directions, clips the active row norms at B_H, reconstructs the complementary covariance by whitened source sampling, and clips its new Gaussian coefficients at B_G. The finite message selecting active inputs has entropy at most D_0; all other new randomness is fresh conditional on the prefix and active matrix. Hence (15) bounds the unprocessed original-weight entropy at stage s by

$$B_s \le NK/t^2 + sD_0 \le 2NK/t^2.$$

For every fixed input x, the conditional reconstruction couplings give expected squared RMS activation injection less than $20e$ per hidden stage, with success indicators. The first layer has the smaller bound e. Before any later row selection, each new internal coefficient has absolute value at most

$$\frac{2B_H\sqrt{q_a+1}}{m} + \frac{\sqrt{2}B_G}{\sqrt{m}}.$$

The readout is truncated at $N\|v\|^2 \le K_0$ and reconstructed using m source samples, with coefficient bound $\sqrt{K_0}/m$ and uniform output error at most $r/16$. The sum of probabilities of rank, Gram, and readout failures is at most $3L\eta$.

For clarity, the quantitative source-reconstruction input is the following estimate. For bounded source activations $|h_j(x)| \le 1$, active dimension at most q_a, and active matrix H_0, its whitened half-uniform, half-leverage sampling law has expected squared RMS error, with the good-Gram indicator, at most

$$\frac{2\|H_0\|_F^2/N + 16(q_a+1)}{m}.$$

The averaged active energy is bounded by V, while the Gram failure probability is at most η. These facts, Gaussian clipping, the active-direction transport estimate, and the squared triangle inequality give the stated $20e$ bound. Each bound holds for every bounded prefix to which the entropy estimate applies. These are the nontrivial reconstruction estimates imported from [4]; the argument below does not infer them merely from an existing compressed network's clean-output accuracy.

B.3 Rounding inside the canonical construction

Choose fixed scalar lattice rounding with error at most ζ and with zero fixed. After first-layer clipping, round all its coefficients before constructing any later prefix. At each internal stage apply the imported reconstruction procedure to the *already rounded* prefix, then round every newly constructed virtual row before processing the next original layer. Finally round the reconstructed readout. Keep the source rows selected at subsequent stages, as in the unrounded construction. There is no terminal rounding pass.

This modification preserves the preceding estimates. Tanh keeps the rounded virtual activations in $[-1,1]$. Each preceding state is specified before the next fresh source list is sampled. Thus the boundedness and conditional-independence assumptions of the sampling estimate remain valid. Rounding new rows is a deterministic function of the same prefix, message, active matrix and fresh Gaussian variables as their unrounded construction. It therefore preserves (15) without adding a message. The resulting prefix may have a discrete law; that law never receives a Gaussian relative-entropy comparison.

Fix an input x. The conditional transports can be glued while keeping the original full-network marginal equal to Q and the outgoing rounded-prefix/future-weight marginal equal to the next canonical stage's law. To do so, extend each transition by the previous conditional law of past original weights given its incoming prefix and remaining original weights. This disintegration preserves all earlier computed prefixes. Define the new postactivation injections by

$$\eta^s = h^s_{\text{rounded virtual}}(x) - \tanh\big(W^s_{\text{original}} h^{s-1}_{\text{rounded virtual}}(x)\big).$$

The final candidate output is exactly the original network's output under these injections and the analogous readout injection. Its marginal law is independent of x, although the coupling may depend on x.

Coefficient rounding changes a first-layer preactivation by at most $(d+1)\zeta$, and every later virtual preactivation by at most $m\zeta$. Both are at most $\sqrt{e}/4$. Consequently the expected squared RMS hidden injection, including rounding, is at most

$$2(20e) + 2(e/16) < 41e.$$

Outside the abort events, Markov's inequality gives

$$\mathbb{P}\left\{\sum_{s<L} \|\eta^s\|^2 / N > r^2/16\right\} \le \frac{656Le}{r^2} = 0.656\alpha.$$

The total abort probability is at most $3L\eta = 0.03\alpha$. The readout rounding error is at most $m\zeta \le \sqrt{e}/4 \le r/64$, where the last inequality uses $\alpha \le 1/8$ and $L \ge 2$. Together with its sampling error, the scalar injection is at most $5r/64$. On the good event,

$$\|\eta\|_* \le \sqrt{r^2/16 + 25r^2/4096} = \frac{\sqrt{281}}{64} r < r/2.$$

The exceptional probability is less than α.

When $N \le M$, the Gaussian entropy-moment inequality instead gives, with probability at least $1 - \alpha/4$, the coefficient bound B_{small} above. On this event round every original coefficient and compute the rounded network sequentially; on its complement use a zero-readout default. At each input the rounded computation is the original network under injections satisfying

$$\|\eta\|_*^2 \le \zeta^2[(d+1)^2 + (L-1)N^2] \le \zeta^2 L(M+d+1)^2 = Le/16 < r^2/4.$$

This identity uses bounded rounded hidden activations, not a bound on products of original matrix norms.

All retained coefficients, after rounding, lie in $[-B_*, B_*]$. Padding each candidate to width M with zero neurons is exact for tanh. A grid with at most $3 + 4B_*/\zeta$ entries per coefficient gives a finite class $\mathcal{H}_*$ with

$$\log|\mathcal{H}_*| \le J_*. \tag{16}$$

The construction proves the following strengthened finite-code statement: for every such Q there is one law Γ_Q on $\mathcal{H}_*$ such that, at each fixed input, a coupling has

$$h(x) = z_w(x;\eta), \qquad \|\eta\|_* < r/2$$

except on probability at most α. Only the coupling, not its two marginals, can depend on the input.

B.4 Posterior and statistical argument

Under the raw Gaussian, NE_v/t^2 is chi-square with N degrees of freedom. The target-cylinder denominator and the exponential moment bound give posterior mass at most β to $E_v > B_v$. The retained conditional posterior Q' satisfies

$$D_{\mathrm{KL}}(Q'\|\mu_{N,t}) \le ND_*/t^2 + \log(1/(1-\beta)) \le NK/t^2.$$

Every retained original network is robust on the common radius r. Apply the strengthened finite-code statement to Q'. On each good coupling the clipped-score error is at most $1/16$.

Let ψ be one on $(-\infty, 1/3]$, zero on $[5/12, \infty)$, and affine between these intervals. A correctly fitted training point has candidate ramp loss zero on the good coupling. A test-margin failure of the original network has candidate ramp loss one there. For the finite class $\mathcal{H}_*$, a relative Chernoff inequality and a union bound give, on a sample event of probability at least $1-\delta$,

$$\mathbb{E}_D\psi(f(X)T(h(X))) \le \frac{2}{n}\sum_{i=1}^{n}\psi(f(X_i)T(h(X_i))) + \frac{8[J_* + \log(1/\delta)]}{n} \qquad (h \in \mathcal{H}_*).$$

Averaging over $\Gamma_{Q'}$, comparing population risks, and restoring the discarded posterior mass bounds the desired risk by

$$\beta + 3\alpha + \frac{8[J_* + \log(1/\delta)]}{n} \le \varepsilon$$

under (13). The finite class and sample event do not depend on N, t. This proves Theorem 5. Adding global-margin conditioning changes none of these steps, since the same target cylinder is retained.

B.5 Width–depth accounting and a sharper target cylinder

For fixed $d, c, \varepsilon, \delta$, the displayed budgets give

$$\begin{aligned} D_*, K, B_v &= O(CLF), & r^{-2} &= O(CL^2F),\\ e^{-1} &= O(CL^3F), & \eta^{-1} &= O(L),\\ q_a, V &= O(C^2L^5F^2), & M &= O(C^3L^8F^3),\\ P_* &= O(C^6L^{17}F^6). \end{aligned}$$

Both B_* and ζ^{-1} are polynomial in C, L, F, so $\log(3 + 4B_*/\zeta) = O(G)$. The old terminal grid had an additional factor L here. This establishes (14).

There is a separate improvement to the target-cylinder budget which changes neither the prior nor the compression. In Appendix A, replace the depth-dependent bulk cutoff by

$$b = \sqrt{16\log 9 + 8\log 4}, \qquad A = \max\{32b, 2\log(8/e_*)\}.$$

Each unconstrained Gaussian bulk matrix has, by the same two-net bound,

$$\mathbb{P}\{\|E^s\|_{\mathrm{op}} > bt\} \le 2\exp(2N\log 9 - Nb^2/8) = 2\cdot 4^{-N} \le 1/2.$$

The bulk matrices are conditionally independent across layers. Their simultaneous success probability is therefore at least $2^{-(H-1)}$. Every deterministic stability estimate in the cylinder proof is unchanged. Recompute B_0, q_0, ρ with this constant-cutoff choice of A and use the conservative cost

$$D_*^{\mathrm{prod}} = B_0 + q_0\log\frac{e\sqrt{2\pi}}{2\rho} + H\log 2. \tag{17}$$

Indeed $N/t^2 \geq 1$ absorbs the product-event cost into this rate. At fixed d, c, with $F_0 = 1+\log^2(2+C)$, one has $D_*^{\mathrm{prod}} = O(CLF_0)$. Substituting this valid cylinder cost in Theorem 5 gives

Corollary 6 (Depth logarithms removed from the target gain)**.** *For fixed $d, c, \varepsilon, \delta$, the unchanged robust prior admits sufficient sample count*

$$O\Big(C^6 L^{17}[1+\log^2(2+C)]^6 \log(2+CL)\Big)$$

at every $N \geq C$ and every $0 < t \leq 1$. At fixed teacher width this is $O(L^{17}\log(2+L))$.

The same cylinder replacement is also available in the quartic main theorem; it is independent of the sampling and quantization argument. Its proof uses the target constants only through their valid mass lower bound. With this substitution the quartic sufficient order becomes

$$O\Big(C^4 L^7[1+\log^2(2+C)]^4 \log^2(2+L)\log(2+CL)\Big).$$

At fixed teacher width this is $O(L^7\log^3(2+L))$.

Remark 7 (A conservative joint polynomial)**.** *Put $S = 2+d+L+C+c+c^{-1}+\varepsilon^{-1}+\log(1/\delta)$. The coarse accounting in [4] for the unchanged budgets gives $e^{-1} = O(S^8)$, $M = O(S^{21})$, $P_* = O(S^{43})$, and $B_* = O(S^{22})$. The new mesh has $\zeta^{-1} = O(S^{25})$. Thus $J_* = O(S^{43}\log S)$ and (13) is $O(S^{44}\log S)$, hence $O(S^{45})$. This deliberately loose degree-45 envelope is a consequence of the conventional modification proved here. It does not replace or enlarge the existing Lean certification of the earlier degree-46 envelope.*

C Bounded-density compression with factored Gaussian reconstruction

The all-scale code in Appendix B can be sharpened in two further ways. First, the actual conditioned posterior has a bounded density relative to the raw Gaussian. Propagating this stronger property through the virtual computation removes the factor $1/\eta$ from the active-direction cutoff. This argument is specific to bounded density; it does not upgrade the compression theorem for arbitrary laws having only a relative-entropy bound. Second, the active part of each reconstructed matrix already has small rank, and its fresh Gaussian part can be replaced by a small-rank Gaussian factorization. Coding the factors then gives a sufficient sample order $\widetilde{O}(C^5L^{12})$ at every $0 < t \leq 1$. The support remains exactly $\mathcal{R}_{N,c}$. This is a supporting code: the entropy-metric construction in the next appendix improves its fifth-power width dependence to the headline quartic bound.

We give the density and factorization arguments below. We use the whole-layer active-direction and whitened-sampling estimates stated in Appendix B, with their original proofs in [4]. As in that appendix, every comparison coupling is allowed to depend on the input, whereas the candidate law is not. The argument and the improved exponents are conventional proofs and are not Lean formalized.

C.1 Explicit budgets and learning statement

Use the product-cylinder cost $D_* = D_*^{\mathrm{prod}}$ from (17). For $0 < \varepsilon, \delta < 1$, define

$$\beta = \varepsilon/4, \qquad B_v = \max\{8, 8[D_* + \log(1/\beta)]\}, \quad K = D_* + 1,$$
$$r = \min\{1/6, c/\sqrt{1+(L-1)B_v}\}, \quad \alpha = \varepsilon/12, \qquad e = \frac{\alpha r^2}{1000L}, \qquad \eta = \frac{\alpha}{100L}. \tag{18}$$

Set

$$q = \left\lceil \frac{2K + \log(1/\eta)}{e} \right\rceil, \qquad D_0 = [(d+1)q + 2]\log 2, \qquad V = 8K + 2q,$$
$$B_H^2 = 4V/e, \qquad K_0 = (4K + 2\log 2)/\eta, \qquad B_1^2 = 4[4K + 2(d+1)\log 2]/e. \tag{19}$$

Let m be the least integer no smaller than each of

$$\begin{aligned} &10000(V + q + 1)/e, \\ &64(q+1)\log(4(q+1)/\eta), \\ &512K_0 r^{-2}[d\log 2 + \log(2/\eta) + 1]. \end{aligned} \tag{20}$$

The factor rank and the small-width cutoff are

$$p = \lceil 1000/e \rceil, \qquad k = q + p, \qquad n_0 = \max\{1, \lceil LD_0/K \rceil\}. \tag{21}$$

In particular, the cutoff n_0 need not be as large as m. Define factor bounds and parameter counts by

$$\begin{aligned} B_g^2 &= 4p/e, \qquad A_f = \max\{B_H, B_g, 1\}, \\ B_f &= \max\{2\sqrt{q+1}/m, \sqrt{2/\eta}, 1\}, \\ P_{\text{small}} &= n_0(d+1) + (L-2)n_0^2 + n_0, \\ P_{\text{fac}} &= m(d+1) + 2(L-2)mk + m, \\ B_{\text{small}}^2 &= \frac{4(4n_0K + 2P_{\text{small}}\log 2)}{\alpha}, \\ B_{\text{code}} &= 1 + \max\{B_{\text{small}}, B_1, A_f, B_f, \sqrt{K_0}/m, 1\}. \end{aligned} \tag{22}$$

Finally, choose the common scalar mesh

$$\zeta = \min\left\{1, \frac{\sqrt{e}}{4(d+1)}, \frac{\sqrt{e}}{4m}, \frac{\sqrt{e}}{4mk(A_f + B_f + 1)}, \frac{\sqrt{e}}{4(n_0 + d + 1)}\right\}, \tag{23}$$

and put

$$J_{\text{fac}} = \log 2 + \max\{P_{\text{small}}, P_{\text{fac}}\}\log(3 + 4B_{\text{code}}/\zeta). \tag{24}$$

All these quantities are independent of learner width and Gaussian scale.

Theorem 8 (All-scale learning by a factored code)**.** *Under the Boolean-teacher convention of the main theorem, the condition*

$$n \geq \frac{16}{\varepsilon}[J_{\text{fac}} + \log(1/\delta)] \tag{25}$$

implies, with probability at least $1 - \delta$ over iid training inputs,

$$\mathbb{E}_{w \sim Q_{S,N,t,c},\, X \sim D} \mathbf{1}\{f(X) z_w(X) \leq 1/4\} \leq \varepsilon$$

simultaneously for every $N \geq C$ and every $0 < t \leq 1$. The same conclusion holds if the posterior conditioning additionally requires the label-independent global margin $|z_w(x)| > 1/2$ for every input x. No population-merging assumption or additional internal-noise condition is imposed.

Corollary 9 (Improved width and depth powers)**.** *For fixed $d, c, \varepsilon, \delta$, let $F_0 = 1 + \log^2(2 + C)$. A sufficient sample count in Theorem 8 has order*

$$O\Big(C^5 L^{12} F_0^5 \log(2 + CL)\Big) = \widetilde{O}(C^5 L^{12}). \tag{26}$$

At fixed teacher width this is $O(L^{12}\log(2+L))$; at fixed depth it is $O(C^5 \log^{11}(2+C))$.

C.2 A bounded-density invariant for unprocessed weights

The following lemma strengthens the entropy bookkeeping at the point where an active rank must hold with high probability. A measure inequality means domination on every measurable set.

Lemma 10 (Conditional density and finite-message propagation)**.** *Suppose that*

$$Q_{A,X,Y} \le e^b(\nu_A \otimes P_X \otimes P_Y),$$

where ν_A, P_X, P_Y are probability reference laws. Write $u = dQ_{A,Y}/d(\nu_A \otimes P_Y)$. Then

$$Q\{u(A,Y) < \eta\} \le \eta,$$

and, outside this event,

$$D_{\mathrm{KL}}(Q_{X|A,Y}\|P_X) \le b + \log(1/\eta). \tag{27}$$

Suppose also that $J = j(A,Y)$ takes values in a fixed finite alphabet $\mathcal{J}$, that $H_0 = h(A,J,X)$, and that a new prefix A' is generated from (A,J,H_0) through a kernel whose fresh randomness has no further dependence on Y. There is a probability reference law $\nu_{A'}$ such that

$$Q_{A',Y} \le e^{b+\log|\mathcal{J}|}(\nu_{A'} \otimes P_Y). \tag{28}$$

The original domination also implies

$$\mathbb{E}_{Q_A} D_{\mathrm{KL}}(Q_{X,Y|A}\|P_X \otimes P_Y) \le b.$$

Proof. Let $F \le e^b$ be the joint density relative to $\nu_A \otimes P_X \otimes P_Y$. Disintegration gives conditional density $F(a,x,y)/u(a,y)$ for $Q_{X|a,y}$ wherever $u > 0$. Thus its relative entropy is at most $b - \log u(a,y)$. Also

$$Q\{u < \eta\} = \int_{\{u<\eta\}} u\, d(\nu_A \otimes P_Y) \le \eta.$$

This is an unconditional probability bound under the canonical joint law; it need not hold separately at every fixed prefix.

For propagation, augment the reference law by an independent uniformly distributed message in $\mathcal{J}$. The actual augmented law satisfies

$$Q_{A,J,X,Y} \le e^b|\mathcal{J}|(\nu_A \otimes \mathrm{Unif}(\mathcal{J}) \otimes P_X \otimes P_Y).$$

Apply the retained-data map and the same prefix kernel to both sides. Under the reference law, its inputs and fresh randomness are independent of Y, so its pushforward factors as $\nu_{A'} \otimes P_Y$. This proves (28). The last assertion follows from the KL chain rule and the nonnegativity of $D_{\mathrm{KL}}(Q_A\|\nu_A)$. □

Suppose initially that a law Q on original weights satisfies

$$Q \le \exp(NK/t^2)\,\mu_{N,t}. \tag{29}$$

The virtual prefix records the constructed network, not just its value at one selected input. Clipping and rounding the first layer are deterministic channels applied to that original block and initialize the domination invariant with exponent NK/t^2. At an internal stage, let X be the current original block and Y the remaining original blocks. Induction with Lemma 10 gives exponent

$$b_s \le NK/t^2 + sD_0 \le 2NK/t^2 \qquad (N > n_0).$$

At that stage abort if its marginal density $u(A, Y)$ is below η. Otherwise (27) and the imported whole-layer active-direction lemma, applied with tolerance e, give

$$\dim U \le \frac{t^2}{Ne}[b_s + \log(1/\eta)] \le \frac{2K + \log(1/\eta)}{e} \le q.$$

The active query list is selected from the conditional law given (A, Y), so it is a function of (A, Y). Its basis is a function of the prefix and the list, using fixed Gram–Schmidt conventions. The selected list and an abort symbol admit a fixed alphabet of log-cardinality at most D_0. For reference messages never chosen by the actual selection rule, extend the basis and transition kernel by fixed measurable default rules. This makes the uniform-message reference construction in the lemma defined on its entire domain.

All abort paths remain in the canonical probability law. On an abort, use a fixed default prefix, retain an unsuccessful flag, and use default transitions thereafter. Do not condition any intermediate law on success. The density-test abort is included in the finite message; later aborts arising from fresh randomness are part of the prefix channel. Consequently the invariant applies on the full law, and the unsuccessful paths are discarded only in the final approximation probability estimate.

The domination invariant also provides the old averaged entropy bound. Thus the imported Gaussian entropy-moment estimate still gives

$$\mathbb{E}\left[\|H_0\|_F^2/N;\ \text{active success}\right] \le 4t^2 b_s/N + 2t^2 q \log 2 \le V. \tag{30}$$

In particular, the definitions of V, B_H, B_1, and K_0 need no extra factor of $1/\eta$. The improvement in q uses the density hypothesis (29); an arbitrary KL bound alone would not justify it.

For comparison, using this new rank cutoff with the dense sequential code of Appendix B already gives $O(C^6 L^{15} F_0^6 \log(2 + CL))$ samples. The factor construction below saves additional powers of both resources.

C.3 A small-rank replacement for the Gaussian field

We recall the source-sampling notation needed for the new step. The source activations obey $|h_j(x)| \le 1$. If U has $\ell \le q$ orthonormal columns, put $u_j = \sqrt{N} U_{j,:}$. The half-uniform, half-leverage law gives weights $a_j = 1/(Np_j) \le 2$ and $a_j \|u_j\|^2 \le 2\ell$. Draw m source indices J_b, and define

$$\widehat{G} = \frac{1}{m}\sum_b a_{J_b} u_{J_b} u_{J_b}^\top, \qquad D_{b,:} = \sqrt{a_{J_b}/m}\, u_{J_b}^\top \widehat{G}^{-1/2}.$$

On the good Gram event, $\widehat{G} \succeq I/2$ and $D^\top D = I$. For $\ell = 0$, the active matrices are empty and the following projection is the identity. Write

$$P = I - DD^\top, \qquad c_b(x) = \sqrt{a_{J_b}/m}\, h_{J_b}(x), \qquad \widehat{C}(x,x) = \|Pc(x)\|^2 \le 2.$$

The ideal field in the imported whitened-sampling estimate has active mean $H_0\psi(x)$ and independent Gaussian rows of variance $t^2 C(x,x)$. The original reconstruction has mean $H_0\widehat{v}(x)$ and variance $t^2\widehat{C}(x,x)$.

Lemma 11 (Factored Gaussian reconstruction)**.** *Conditional on the source samples and retained active data, draw an independent standard Gaussian matrix $R \in \mathbb{R}^{p\times m}$ and independent standard Gaussian row vectors $g_i \in \mathbb{R}^p$, for all virtual target rows. Replace the reconstructed coefficients by*

$$\widetilde{W}_{ib} = \frac{a_{J_b}}{m} H_{0,i} \cdot u_{J_b} + t\sqrt{\frac{a_{J_b}}{mp}}\,(g_i^\top RP)_b. \tag{31}$$

This coefficient law does not depend on the input. At every fixed input it can be coupled to the ideal field so that the expected squared RMS evaluation error, with the good Gram indicator, is at most

$$\frac{4\left\|H_0\right\|_F^2/N + 32(q+1)}{m} + \frac{8}{p}. \tag{32}$$

The coefficient matrix in (31) has rank at most $\ell + p$.

Proof. Fix the input and abbreviate $\sigma^2 = \widehat{C}(x,x)$. Conditional on R, the new Gaussian evaluations are independent across rows with variance $t^2 s^2$, where

$$s = \left\|RPc(x)\right\|/\sqrt{p}.$$

If $\sigma > 0$, then s^2/σ^2 has the law of a chi-square variable with p degrees of freedom divided by p. Rationalizing the square root therefore gives

$$\mathbb{E}_R(s-\sigma)^2 \le \sigma^2 \mathbb{E}(\chi_p^2/p - 1)^2 = 2\sigma^2/p \le 4/p. \tag{33}$$

For $\sigma = 0$ the statement is immediate.

To make the coupling explicit, draw R and independent standard normals Z_i, independent of R. Use Gaussian row evaluations $t\sigma Z_i$ for the old reconstruction and tsZ_i for the new one. Disintegrating g_i conditional on its projection along $RPc(x)$ realizes the latter evaluation while keeping its full conditional marginal standard Gaussian. Thus the coefficient law is exactly the law in (31). Disintegration likewise extends the old Gaussian evaluation to its coefficient field. The degenerate zero-variance case is handled by independent completion. Only this comparison coupling, not either field marginal, depends on x.

The old whitened reconstruction estimate is $[2\left\|H_0\right\|_F^2/N + 16(q+1)]/m$. Couple through that reconstruction and apply the squared triangle inequality with (33). This yields (32). Finally the active term factors through $\mathbb{R}^\ell$ and the Gaussian term through $\mathbb{R}^p$, proving the rank claim. □

Integrating (32) with the success indicator and using (30), the first line of (20), and $p \ge 1000/e$, bounds this error by

$$\frac{4V + 32(q+1)}{m} + \frac{8}{p} < e/8.$$

The comparison requires no simultaneous coupling of all inputs. Conditional independence of rows given R suffices; independence of the new rows after averaging over R is not asserted or needed.

C.4 Bounded factors and their sequential rounding

At each internal stage first clip active rows of norm greater than B_H to zero, exactly as in the imported construction. Equation (30) and bounded activations give squared RMS activation cost at most $4V/B_H^2 = e$. Apply Lemma 11 to this clipped matrix.

Abort if $\left\|R\right\|_F^2 > pm/\eta$. Since $\mathbb{E}\left\|R\right\|_F^2 = pm$, this event has conditional probability at most η. Also replace each g_i by zero when $\left\|g_i\right\|^2 > B_g^2 = 4p/e$. This row replacement has probability at most $e/4$ and hence expected squared activation change at most e: two tanh values differ by at most two. All virtual rows are clipped before any later row selection, including selections depending on future weights.

On success, write the resulting coefficients as $A_i B$, with zero padding to factor dimension $k = q + p$, where

$$A_i = (H_{0,i}, g_i), \qquad B_{\text{active},ab} = a_{J_b} u_{J_b,a}/m,$$

$$B_{\mathrm{Gaussian},ab} = t\sqrt{a_{J_b}/(mp)}\,(RP)_{ab}.$$

Every left factor entry has absolute value at most A_f. Every right factor entry has absolute value at most B_f, since

$$|a_{J_b}u_{J_b,a}/m| \le 2\sqrt{q+1}/m, \qquad t\sqrt{\frac{a_{J_b}}{mp}}|(RP)_{ab}| \le \sqrt{\frac{2}{mp}}\,\|R\|_F \le \sqrt{2/\eta}.$$

Round both factor arrays to the common scalar mesh (23) before computing the next virtual prefix. For each bounded incoming state,

$$|(A_{i,\mathrm{round}}B_{\mathrm{round}} - A_iB)h| \le mk(A_f + B_f + 1)\zeta \le \sqrt{e}/4. \tag{34}$$

Here each product changes by at most $(A_f + B_f + 1)\zeta$, because $\zeta \le 1$. Round first-layer coefficients and reconstructed readout coefficients directly. Their preactivation errors are at most $(d+1)\zeta$ and $m\zeta$, respectively, again at most $\sqrt{e}/4$. The rounded candidate is an ordinary depth-L tanh network: its internal matrix is the product of two rounded factor arrays. Factorization adds no activation layer.

All source samples, R, and Gaussian row factors are fresh randomness conditional on the prefix, query list, and retained active matrix. Clipping, factor multiplication, and rounding are deterministic subsequent operations. They therefore form a channel of the precise kind required by Lemma 10. The reference-prefix law need not be Gaussian or have a density on its own coordinate space. The comparison coupling does not replace this canonical channel in the density induction.

C.5 The disturbance budget and the finite class

Consider first $N > n_0$ and a law satisfying (29). The first layer is clipped at the physical standardized row bound B_1 and then rounded. Its expected squared RMS activation injection is less than $3e$ by the imported first-layer estimate and the squared triangle inequality. At an internal stage the five contributions are active-direction transport, active-row clipping, reconstruction, Gaussian-row clipping, and factor rounding. Their expected squared RMS bounds, with success indicators, are respectively

$$2e, \quad e, \quad e/8, \quad e, \quad e/16.$$

Another squared triangle inequality bounds their sum by $5(2 + 1 + 1/8 + 1 + 1/16)e < 21e$. In particular the conservative bound $41e$ from Appendix B remains available.

The good Gram probability follows from the second line of (20). Use the original readout construction: truncate at $N\|v\|^2 \le K_0$, sample source indices with probabilities proportional to $|v_i|$, and use signed coefficients of magnitude $\|v\|_1/m \le \sqrt{K_0}/m$. The last line of (20) gives output sampling error at most $r/16$ uniformly over the finite input domain except with probability η. The energy truncation also fails with probability at most η. Including density-test, Gram, and shared-sketch aborts at internal stages, the total abort probability is at most $4L\eta$.

For each fixed input, glue the conditional transports as in Appendix B: retain the full original-weight marginal Q and the canonical outgoing prefix/future-weight marginal at each transition. Extend the transition by the previous conditional law of past original weights. This preserves every earlier computed prefix. The final candidate output is exactly the original network under the postactivation and readout injections defined by its rounded virtual computation. No robustness of an intermediate candidate is assumed.

On the non-aborted paths, Markov's inequality gives

$$\mathbb{P}\left\{\sum_{s<L}\|\eta^s\|^2/N > r^2/16,\ \text{no abort}\right\} \le \frac{656Le}{r^2} = 0.656\alpha.$$

Readout rounding adds at most $\sqrt{e}/4 \le r/64$. Thus, outside probability at most $0.656\alpha + 4L\eta < \alpha$, the total disturbance satisfies

$$\|\eta\|_* \le \sqrt{r^2/16 + (r/16 + r/64)^2} = \frac{\sqrt{281}}{64} r < r/2.$$

The construction above does not require $m \le N$. Source sampling is with replacement, and duplicated indices give duplicated stored neurons. The Gram lemma also allows this: on its good event D has orthonormal columns, and $\ell \le q < m$. Hence the only width cutoff needed for this branch is $N > n_0$, which pays for the accumulated message exponents.

For $N \le n_0$, the original density bound implies a KL bound NK/t^2. The Gaussian entropy-moment inequality gives, for all physical standardized coordinates together,

$$\mathbb{E}_Q \|tZ_{\text{all}}\|^2 \le 4n_0K + 2P_{\text{small}} \log 2.$$

With probability at least $1 - \alpha/4$, every actual coefficient is at most B_{small} in absolute value. On that event round the original network directly; otherwise use a zero-output default. The bounded rounded activations give the exact injection bound

$$\|\eta\|_*^2 \le \zeta^2[(d+1)^2 + (L-1)N^2] \le \zeta^2 L(n_0 + d + 1)^2 \le Le/16 < r^2/4.$$

Pad these small networks only to width n_0.

The large branch keeps the m selected rows at each layer. It stores $m(d+1)$ first-layer scalars, $2mk$ factor scalars for each internal matrix, and m readout scalars. The small branch stores at most P_{small} scalars. Every rounded scalar in either branch belongs to a fixed alphabet of at most $3 + 4B_{\text{code}}/\zeta$ values. The union of these two fixed finite classes, denoted $\mathcal{H}_{\text{fac}}$, therefore satisfies

$$\log|\mathcal{H}_{\text{fac}}| \le J_{\text{fac}}. \tag{35}$$

In particular one does not count the small branch as a dense network of width m.

We have proved that every law satisfying (29) admits a single candidate law Γ_Q on $\mathcal{H}_{\text{fac}}$ with the following property: at each fixed input it has a coupling with $w \sim Q$ and a disturbance such that

$$h(x) = z_w(x;\eta), \qquad \|\eta\|_* < r/2 \tag{36}$$

except with probability at most α. Both marginals are independent of the input used to form the comparison coupling.

C.6 Posterior comparison and polynomial accounting

Proof of Theorem 8. The product target cylinder gives raw conditioning probability at least $\exp(-ND_*/t^2)$. The chi-square exponential moment for the readout, combined with this denominator, gives

$$Q_{S,N,t,c}\{E_v > B_v\} \le \exp\left(-\frac{NB_v}{4t^2} + \frac{N}{2}\log 2 + \frac{ND_*}{t^2}\right) \le \beta.$$

Let Q' be the posterior conditioned on $E_v \le B_v$. Unlike a generic law controlled only in relative entropy, it satisfies the literal density bound

$$\frac{dQ'}{d\mu_{N,t}} \le \frac{\exp(ND_*/t^2)}{1-\beta} \le \exp(NK/t^2).$$

Every original network in its support is robust on the common radius r. Consequently (36) implies clipped-score error at most $1/16$ on its good event.

Use the ramp ψ from the earlier statistical proof: it is one on $(-\infty, 1/3]$, zero on $[5/12, \infty)$, and affine in between. Every training point has candidate ramp loss zero on the good coupling; every original test-margin failure has candidate loss one there. A relative Chernoff bound and a union bound over $\mathcal{H}_{\mathrm{fac}}$ give, on one sample event of probability at least $1-\delta$,

$$\mathbb{E}_D \psi(f(X)T(h(X))) \le \frac{2}{n}\sum_{i=1}^{n} \psi(f(X_i)T(h(X_i))) + \frac{8[J_{\mathrm{fac}} + \log(1/\delta)]}{n} \quad (h \in \mathcal{H}_{\mathrm{fac}}).$$

Averaging this inequality under $\Gamma_{Q'}$, applying the two marginal comparisons, and restoring the discarded readout tail bounds the desired risk by

$$\beta + 3\alpha + \frac{8[J_{\mathrm{fac}} + \log(1/\delta)]}{n} \le \varepsilon.$$

The finite class and sample event do not depend on N, t, which gives the simultaneous conclusion. Additional global clean-margin conditioning retains the same target cylinder, density bound, and readout-tail argument, so the same proof applies to that stated optional variant. □

Proof of Corollary 9. The product-cylinder constants give, at fixed ancillary resources,

$$K, B_v = O(CLF_0), \qquad r^{-2} = O(CL^2F_0), \qquad e^{-1} = O(CL^3F_0), \qquad \eta^{-1} = O(L).$$

The term $\log(1/\eta)$ in the rank bound is dominated by the target cost K. Therefore

$$q, V = O(C^2L^4F_0^2), \qquad p = O(CL^3F_0), \qquad m = O(C^3L^7F_0^3), \qquad n_0 = O(CL^4F_0).$$

The Gram and readout terms in the maximum defining m, including their logarithms, are dominated by its first term at this fixed-resource scale. It follows that

$$P_{\mathrm{fac}} = O(C^5L^{12}F_0^5), \qquad P_{\mathrm{small}} = O(C^2L^9F_0^2).$$

All coefficient bounds and inverse meshes are polynomial in C, L, F_0, so $\log(3 + 4B_{\mathrm{code}}/\zeta) = O(\log(2+CL))$. Substitution into (24) and (25) proves the claimed order and both special cases. □

Remark 12 (A conservative joint polynomial for the factored code)**.** *Put*

$$S = 2 + d + L + C + c + c^{-1} + \varepsilon^{-1} + \log(1/\delta).$$

The target-cylinder estimates and (18) *give*

$$K, B_v = O(S^3), \qquad r^{-2} = O(S^6), \qquad e^{-1} = O(S^8), \qquad \eta^{-1} = O(S^2).$$

The new budgets consequently satisfy

$$q, V = O(S^{11}), \quad D_0 = O(S^{12}), \quad m = O(S^{19}), \quad n_0 = O(S^{13}), \quad p = O(S^8),$$

$$P_{\mathrm{fac}} = O(S^{31}), \quad P_{\mathrm{small}} = O(S^{27}), \quad B^2_{\mathrm{small}} = O(S^{28}), \quad B_{\mathrm{code}} = O(S^{14}).$$

For the mesh, $A_f = O(S^{10})$ and $B_f = O(S)$; the latter uses the first lower bound on m to bound $\sqrt{q+1}/m$ by a constant. Thus $\zeta^{-1} = O(S^{44})$, where the factor-rounding denominator dominates the other displayed choices. It follows that

$$J_{\mathrm{fac}} = O(S^{31}\log S), \qquad n = O(S^{32}\log S) \subseteq O(S^{33}).$$

This degree-33 envelope is deliberately conservative. Like the refined width and depth powers above, it is a consequence of the present conventional argument and carries no Lean-certification claim.

D Entropy-weighted arithmetic compression

This appendix constructs a finite comparison class while leaving the original Gaussian prior and the robustness support unchanged. The comparison functions use a finite arithmetic decoder between tanh layers; they need not be ordinary neural networks. This is sufficient for the finite-class statistical argument and for the layerwise internal-noise coupling. The result is a conventional proof, not a Lean formalization.

D.1 A single entropy matrix for every Gaussian channel

Lemma 13 (Gaussian-channel entropy matrix)**.** *Let P be a probability law on $\mathbb{R}^{N\times q}$ with finite $D = D_{\mathrm{KL}}(P\|\gamma_{Nq})$. There is a positive semidefinite $q\times q$ matrix T with $\operatorname{tr}\mathsf{T} = D$ such that, for*

$$Q_b = \mathrm{Law}(Xb + \sqrt{1-\|b\|^2}\,Z), \qquad X\sim P, \quad Z\sim\gamma_N \text{ independent},$$

every $\|b\|<1$ and $\|b'\|\le 1$ satisfy

$$W_2^2(Q_b,Q_{b'}) \le \frac{8(b'-b)^\top \mathsf{T}(b'-b)}{1-\|b\|^2}. \tag{EM1}$$

The same matrix works for all channel directions.

Proof. Let P_s be the Ornstein–Uhlenbeck evolution of P, let $f_s = dP_s/d\gamma_{Nq}$, and view its relative score as an $N\times q$ matrix. Define

$$\mathsf{I}_s = \mathbb{E}_{P_s}[(\nabla\log f_s)^\top(\nabla\log f_s)], \qquad \mathsf{T} = \int_0^\infty \mathsf{I}_s\,ds.$$

Gaussian entropy dissipation gives $\operatorname{tr}\mathsf{T} = D$. Score commutation and conditional Jensen give, in Loewner order,

$$\mathsf{I}_t \preceq e^{-2(t-s)}\mathsf{I}_s \quad (0<s<t), \qquad \mathsf{I}_t \preceq \frac{2}{e^{2t}-1}\mathsf{T}. \tag{EM2}$$

First assume a smooth positive density. Put $C = 1-\|b\|^2$. For a direction velocity δ, differentiating $Y = Xb+\sqrt{C}Z$ yields the continuity-equation velocity

$$v_b(y) = \mathbb{E}\left[X\delta - \frac{b\cdot\delta}{\sqrt{C}}Z \mid Y=y\right] = \mathbb{E}[\nabla\log f(X)\mid Y=y]\,\delta. \tag{EM3}$$

To verify the equality, Gaussian integration by parts under the reference conditional law of X given Y gives

$$\mathbb{E}[X\mid Y] - Yb^\top = \mathbb{E}[\nabla\log f(X)\mid Y](I-bb^\top).$$

Substitution, using $(I - bb^\top)^{-1} = I + bb^\top/C$, proves (EM3).

For $\rho = \|b\|^2 > 0$, take t such that $e^{-2t} = \rho$. The channel is a projection of the full observation $X_t = \sqrt{\rho}X + \sqrt{1-\rho}G$. Score commutation and (EM2) imply

$$\mathbb{E}\|v_b(Y)\|^2 \le \rho^{-1}\delta^\top \mathsf{I}_t \delta \le \frac{2}{C}\delta^\top \mathsf{T}\delta. \tag{EM4}$$

The case $b = 0$ follows by continuity. Along the straight path $b_s = (1-s)b + sb'$, with $\delta = b' - b$, the residual variance is

$$C_s = (1-s)C + s(1 - \|b'\|^2) + s(1-s)\|\delta\|^2 \ge (1-s)C.$$

The Wasserstein length bound and (EM4) give

$$W_2(Q_b, Q_{b'}) \le \sqrt{2\delta^\top \mathsf{T}\delta}\int_0^1 C_s^{-1/2}\, ds \le 2\sqrt{2\delta^\top \mathsf{T}\delta/C}.$$

For general finite-entropy P, apply the result first to P_ϵ. Its entropy matrix is $\int_\epsilon^\infty \mathsf{I}_s ds \preceq \mathsf{T}$. Finite Gaussian relative entropy implies a finite second moment; OU smoothing and the channel outputs converge in W_2 as $\epsilon \downarrow 0$. This proves the stated generality and also the boundary endpoint $\|b'\| = 1$. □

For a physical active matrix $H = tX$, use

$$T = \frac{t^2}{N}\mathsf{T}.$$

If $\|b\|^2 + C = V$, $C > 0$, and $\|b'\|^2 \le V$, rescaling (EM1) gives the following squared RMS bound for the two physical channels:

$$\frac{1}{N}W_2^2\left(\mathrm{Law}(Hb + t\sqrt{C}Z), \mathrm{Law}(Hb' + t\sqrt{V - \|b'\|^2}Z')\right) \le \frac{8V}{C}(b'-b)^\top T(b'-b). \tag{EM5}$$

In our construction $V \le 2$.

D.2 Budgets and the finite message

Fix $K \ge 1$, $0 < \alpha \le 1/8$, $0 < r \le 1/6$, $L \ge 2$, and a scale ceiling $0 < t_0 \le 1$. Set

$$\begin{gathered} e = \frac{\alpha r^2}{1000L}, \qquad \eta = \frac{\alpha}{100L}, \\ K_a = 2K + \log(1/\eta), \qquad q = \lceil K_a/e \rceil, \qquad B = 8K_a. \end{gathered} \tag{EM6}$$

At a successful internal stage the bounded-density argument and the whole-layer active-direction lemma give rank at most q and a conditional physical entropy matrix with $\mathrm{tr}\, T \le K_a$. Round its symmetric entries to the nearest multiple of e/q and add $2eI$. The resulting matrix $\overline{T}$ satisfies

$$\overline{T} \succeq T + eI, \qquad \mathrm{tr}\,\overline{T} \le B, \qquad \kappa(\overline{T}) \le B/e. \tag{EM7}$$

Indeed, the rounding error has operator norm at most $e/2$, and $qe \le K_a + e \le 2K_a$. Smaller active ranks are padded by zeros before this operation, with a separate diagonal floor on the unused block.

Let

$$\begin{gathered} A_T = 3 + 8K_a q/e, \qquad M = \max\{1, 2LD_0/K\}, \\ D_0 = [(d+1)q + 2]\log 2 + q^2 \log A_T. \end{gathered} \tag{EM8}$$

The query list, abort flag, and quantized matrix form a fixed finite message of log-cardinality at most D_0. Invalid reference matrix messages use a fixed positive definite default and a fixed transition.

For a law

$$Q \le \exp(NK/t^2)\mu_{N,t}, \qquad 0 < t \le t_0, \tag{EM9}$$

the future-density invariant has exponent $b_s \le NK/t^2 + sD_0$. In the branch $N/t^2 > M$, this is at most $2NK/t^2$. The density test discards probability at most η at each stage and bounds conditional current-block entropy by $2NK/t^2 + \log(1/\eta)$. This justifies both q and (EM7). The matrix message depends on the prefix and future original weights; after it is supplied the reconstruction kernel has no additional dependence on those future weights.

D.3 Independent-batch regression in the entropy metric

At a stage, let $\ell \le \min(q, N)$ be its actual active rank. All regression and projection formulas in this subsection and the next operate on the ℓ-dimensional active block; q remains only the budget, mesh, and counting cap. In these formulas $\overline{T}$ denotes the corresponding active block. Coordinates are padded only for final storage, with a separate diagonal metric on the unused block, so that the padded projection keeps all unused coordinates zero. Write $u_j = \sqrt{N}U_{j,:}$ for the active source features and, at a fixed input, put

$$\psi = \frac{1}{N}\sum_j u_j h_j, \quad K_x = \frac{1}{N}\sum_j h_j^2, \quad R_j = h_j - u_j^\top \psi, \quad C = \frac{1}{N}\sum_j R_j^2.$$

Then $N^{-1}\sum_j u_j u_j^\top = I_\ell$, $N^{-1}\sum_j u_j R_j = 0$, and $K_x = \|\psi\|^2 + C \le 1$.

Choose

$$m_0 = \lceil \max\{16q, 64(q+1)\log(4(q+1)/\eta)\} \rceil, \qquad m_A = \lceil 10^5 B/e \rceil. \tag{EM10}$$

Use m_0 ordinary half-uniform, half-leverage samples for a pilot $\psi_0 = G_0^{-1} v_0$. Abort if $G_0 \not\succeq I/2$. The usual matrix Chernoff bound makes this probability at most η. With the good-pilot indicator,

$$\mathbb{E}\|\psi_0 - \psi\|^2 \le 8qC/m_0 \le C/2. \tag{EM11}$$

Independently draw m_A samples with probabilities

$$p_j = \frac{1}{3N}\left(1 + \frac{\|u_j\|^2}{\ell} + \frac{u_j^\top \overline{T} u_j}{\operatorname{tr}\overline{T}}\right), \qquad a_j = (Np_j)^{-1}. \tag{EM12}$$

Empty active ranks use uniform sampling and empty matrices. In positive rank, (EM12) gives

$$a_j \le 3, \qquad a_j\|u_j\|^2 \le 3q, \qquad a_j u_j^\top \overline{T} u_j \le 3 \operatorname{tr}\overline{T}.$$

Define

$$\widehat{\psi} = \psi_0 + \frac{1}{m_A}\sum_b a_{J_b} u_{J_b}(h_{J_b} - u_{J_b}^\top \psi_0). \tag{EM13}$$

Conditional on the pilot this estimator is unbiased for ψ. Since $N^{-1}\sum_j (h_j - u_j^\top\psi_0)^2 = C + \|\psi_0 - \psi\|^2$,

$$\mathbb{E}[\|\widehat{\psi} - \psi\|_{\overline{T}}^2; \text{good pilot}] \le \frac{3BC}{m_A}\left(1 + \frac{8q}{m_0}\right) \le \frac{9BC}{2m_A}. \tag{EM14}$$

This calculation does not assume the false isotropic residual bound $\mathbb{E}bb^\top \preceq 2CI/m$. The independent correction is what avoids conjugating the entropy metric by a random Gram inverse. When $C = 0$, the pilot and corrected estimator are both exactly ψ on the good pilot event.

The estimator is linear in its sampled activation values. Write it as $\widehat{\psi}(x) = Fh_{\mathrm{src}}(x)$, with zero columns for unused source groups. A deterministic entry bound on success is

$$|F_{ab}| \le F_* := 20(q+1)^2. \tag{EM15}$$

For example, pilot columns have norm at most $4\sqrt{q}/m_0$; the second empirical Gram has norm at most $3q$; and correction columns have norm at most $3\sqrt{q}/m_A$. These facts imply (EM15).

D.4 A positive variance envelope and the canonical row law

Set

$$\lambda = \min\{1, e/(10^5 t_0^2)\}, \qquad m_K = \max\left\{1, \left\lceil \frac{\log(2/\eta)}{2\lambda^2} \right\rceil\right\}. \tag{EM16}$$

Take a third independent uniform source batch and define

$$\widehat{K}(x) = \frac{1}{m_K}\sum_b h_{J_b}(x)^2, \qquad K_+(x) = \widehat{K}(x) + \lambda.$$

At each fixed input, Hoeffding gives

$$\Pr\{|\widehat{K} - K_x| > \lambda\} \le \eta. \tag{EM17}$$

For $\lambda = 1$ the event is empty. On its complement, $K_x \le K_+ \le K_x + 2\lambda$, and always $0 < K_+ \le 2$. This input-specific event is discarded only in the comparison; the canonical function is defined at every input.

Define the metric projection

$$a(x) = \operatorname*{argmin}_{\|z\|^2 \le K_+(x)} \|z - \widehat{\psi}(x)\|_{\overline{T}}^2, \qquad \sigma(x) = \sqrt{K_+(x) - \|a(x)\|^2}. \tag{EM18}$$

The minimizer is unique. On the event in (EM17), ψ is feasible; nonexpansiveness of Hilbert-space projection gives

$$\|a - \psi\|_{\overline{T}}^2 \le \|\widehat{\psi} - \psi\|_{\overline{T}}^2. \tag{EM19}$$

The unrounded virtual target row is

$$h_i^{\mathrm{new}}(x) = \tanh\{H_i a(x) + t g_i \sigma(x)\}, \qquad g_i \sim N(0,1) \text{ independently over rows.} \tag{EM20}$$

One scalar Gaussian per row suffices. Its cross-input covariance need not match the original field: every comparison is at one fixed input, where (EM20) has the required independent Gaussian row marginals. The complete function law in (EM20) is nevertheless input-independent.

To bound its error, put $\Delta = K_+ - K_x \in [0, 2\lambda]$ on (EM17). Increasing the ideal residual variance from C to $C + \Delta$ costs at most $t^2\Delta \le 2t_0^2\lambda$ in squared RMS. The enlarged ideal field and (EM20) have common Gaussian-reference variance $K_+ \le 2$. For $C > 0$, (EM5), (EM19), and (EM14) therefore give, after the squared triangle inequality and averaging,

$$\mathbb{E}[\text{squared RMS reconstruction error}; \text{good}] \le 4t_0^2\lambda + \frac{144B}{m_A} < e/8. \tag{EM21}$$

The inequality uses $T \preceq \overline{T}$ and $C + \Delta \ge C$. For $C = 0$, exact interpolation makes $a = \psi$, and only the added variance remains. Thus (EM21) also covers that case.

The channel coupling need not preserve the realized H. Conditional on prefix and future weights, both sides have the same specified active-matrix marginal. Disintegration extends the evaluation coupling to the original current block on one side and to the complete canonical candidate field on the other. This preserves the outgoing prefix/future marginal required by the sequential gluing argument.

D.5 Clipping, an explicit mesh, and the decoder size

Retain the old first-layer and readout budgets, and set

$$V_H = 8K + 2q, \quad B_H^2 = 4V_H/e, \quad B_Z = 2/\sqrt{e}, \quad B_1^2 = 4[4K + 2(d+1)\log 2]/e,$$

$$\begin{aligned} K_0 = (4K + 2\log 2)/\eta, \qquad m = m_0 + m_A + m_K + m_R, \\ m_R = \left\lceil 512 K_0 r^{-2}[d\log 2 + \log(2/\eta) + 1] \right\rceil. \end{aligned} \tag{EM22}$$

Clip an active row to zero when its norm exceeds B_H, and clip the physical scalar tg_i to zero when its absolute value exceeds B_Z. The old entropy-moment estimate makes the active clipping cost at most e in expected squared activation RMS. Since $\Pr\{|tg_i| > B_Z\} \le e/4$, Gaussian clipping has the same bound. Both clippings occur for all virtual rows before any future row selection. The entropy matrix is computed from the unclipped active law; clipping is paid after the comparison in (EM21).

Put

$$\begin{aligned} A_* = \sqrt{B/e}\, m\sqrt{q+1}, \qquad U_* = 2\sqrt{q+1} + B_H A_* + 2, \\ \zeta = \min\left\{1, \frac{\sqrt{e}}{8U_*}, \frac{e}{256 B_Z^2 A_*}, \frac{\sqrt{e}}{4(d+1)}, \frac{\sqrt{e}}{4m}\right\}. \end{aligned} \tag{EM23}$$

Round the retained active scalars, entries of F, and physical Gaussian scalars to this common mesh. The matrix $\overline{T}$ stays on its defining finite grid. The current incoming activations are already those of the rounded canonical prefix, so $\widehat{K}$ and the projection radius are unchanged when performing this local rounding comparison.

Rounding F changes its feature vector by at most $m\sqrt{q+1}\zeta$. Projection is $\sqrt{B/e}$-Lipschitz in Euclidean norm, so its output changes by at most $A_*\zeta$. Both projected vectors have norm at most $\sqrt{2}$, and hence

$$|\sigma(a) - \sigma(a')|^2 \le |\|a\|^2 - \|a'\|^2| \le 3\|a - a'\|.$$

For every bounded incoming state, the complete preactivation rounding error is therefore at most

$$U_*\zeta + 2B_Z\sqrt{A_*\zeta} \le \sqrt{e}/4. \tag{EM24}$$

First-layer and readout rounding have the same bound from the last two terms of (EM23). All meshes and coefficient bounds are polynomial in the theorem resources.

The decoder computes Fh, the average of the designated source squares, the metric projection (EM18), the scalar square root, and then (EM20). The projection is an exact operation of this finite arithmetic decoder; no computational-complexity assertion is needed. Let

$$\begin{aligned} B_{\mathrm{code}} = 1 + \max\{B_1, B_H, B_Z, F_*, \sqrt{K_0}, 1\}, \\ P = m(d+1) + (L-2)(2mq + m) + m, \\ J_{\mathrm{EM}} = \log 2 + P\log(3 + 4B_{\mathrm{code}}/\zeta) + (L-2)q^2 \log A_T. \end{aligned} \tag{EM25}$$

Padding unused source groups and rows gives a single finite function class $\mathcal{H}_{\mathrm{EM}}$ with $\log|\mathcal{H}_{\mathrm{EM}}| \le J_{\mathrm{EM}}$, independent of N and of the actual scale $0 < t \le t_0$.

D.6 Compression and resource consequences

Theorem 14 (Arithmetic compression)**.** *With (EM6)–(EM25), suppose (EM9) and $N/t^2 > M$. There is one law Γ_Q on $\mathcal{H}_{\mathrm{EM}}$ such that, at every fixed Boolean input, it can be coupled to the original law Q and an internal disturbance ξ satisfying*

$$h(x) = z_w(x;\xi), \qquad \|\xi\|_* < r/2$$

except with probability at most α. The two specified marginals do not depend on the input used for the comparison.

Proof. The active-direction transport, active clipping, reconstruction, Gaussian clipping, and rounding contributions have expected squared RMS at most $2e, e, e/8, e, e/16$, respectively. The squared triangle inequality retains the old conservative $41e$ bound per hidden stage. The first-layer bound is less than $3e$. Density tests and pilot Gram failures are canonical aborts; variance-envelope failures are input-specific comparison failures. Include every abort path in the canonical law and use default transitions thereafter. Along with the old readout truncation and sampling failures their total probability is at most $4L\eta$.

Every new-prefix operation uses only the prefix, finite message, retained original active matrix, and fresh randomness. Thus the bounded-density induction is valid on the full law. The conditional evaluation couplings glue while preserving the original-weight marginal and each outgoing canonical prefix/future marginal. The original network evaluated under the resulting postactivation injections has exactly the candidate output. No robustness of a virtual prefix is used.

The old disturbance calculation now applies unchanged:

$$\Pr\left\{\sum_{s<L} \|\xi^s\|^2/N > r^2/16, \text{ no comparison failure}\right\} \le 656Le/r^2 = 0.656\alpha.$$

The readout approximation and rounding contribute at most $r/16 + r/64$. Consequently the total disturbance is less than $r/2$ outside probability $0.656\alpha + 4L\eta < \alpha$. □

The complementary branch $N/t^2 \le M$ should be handled by the raw-prior posterior generalization bound, whose relative-entropy charge is at most $NK/t^2 \le MK = O(LD_0)$. A dense parameter code in that branch would lose the improvement. Uniformity in width and scale is supplied by the separate finite Gaussian-scale domination argument. Since $m \ge q$, this charge and (EM25) are both bounded by the coarser expression $\widetilde{O}(L[q^2 + (d+q)m])$.

Ignoring only logarithmic factors and fixed ancillary resources,

$$q = \widetilde{O}(K/e), \qquad m = \widetilde{O}\left(K/e + t_0^4/e^2 + m_R\right),$$

$$J_{\mathrm{EM}} + LD_0 = \widetilde{O}\left(LK^2/e^2 + t_0^4 LK/e^3\right). \tag{EM26}$$

Here the displayed orders also assume the original readout count is dominated by $\widetilde{O}(K/e)$, as it is for the posterior budgets under consideration. At $t_0 = 1$, the product-cylinder bounds $K = O(CLF_0)$ and $e^{-1} = O(CL^3F_0)$ give

$$J_{\mathrm{EM}} + LD_0 = \widetilde{O}(C^4L^{11}F_0^4).$$

The original ramp-loss comparison applies to this finite function class, using the same posterior readout truncation and common robustness radius.

E Direct posterior bounds and interpolation between Gaussian scales

The finite message used by the entropy-metric reconstruction has a quadratic dependence on the active rank. Its accumulated cost need not be converted into a dense small-width code. The actual conditioning posterior can be treated directly whenever that message cost is larger than its Gaussian relative-entropy bound. We give a single statistical event for this argument and for the scale-dependent reconstruction. All reference laws below are fixed before the training sample is drawn.

Lemma 15 (A relative-entropy risk inequality)**.** *Let Π be a probability law on a measurable family of predictors, and let $\ell_h(x) \in [0,1]$. Write $R(h) = \mathbb{E}_X \ell_h(X)$ and $\widehat{R}_S(h) = n^{-1}\sum_{i=1}^n \ell_h(X_i)$. With sample probability at least $1-\delta$, simultaneously for every probability law $\Gamma \ll \Pi$,*

$$\mathbb{E}_\Gamma R \le 2\mathbb{E}_\Gamma \widehat{R}_S + \frac{2}{n}\left[D_{\mathrm{KL}}(\Gamma\|\Pi) + \log(1/\delta)\right]. \tag{37}$$

Proof. Put $a = 1 - e^{-1} > 1/2$. Convexity gives $e^{-u} \le 1 - au$ for $0 \le u \le 1$. For each fixed predictor, independence of the sample therefore gives

$$\mathbb{E}_S \exp\{n[aR(h) - \widehat{R}_S(h)]\} \le e^{naR(h)}[1 - aR(h)]^n \le 1.$$

Fubini and Markov's inequality imply, with probability at least $1-\delta$,

$$\int \exp\{n[aR(h) - \widehat{R}_S(h)]\}\, d\Pi(h) \le 1/\delta.$$

On this event, Jensen's inequality applied under Γ gives

$$n\left[a\mathbb{E}_\Gamma R - \mathbb{E}_\Gamma \widehat{R}_S\right] \le D_{\mathrm{KL}}(\Gamma\|\Pi) + \log(1/\delta).$$

Indeed, the general inequality used here is $\int F\, d\Gamma - D_{\mathrm{KL}}(\Gamma\|\Pi) \le \log \int e^F\, d\Pi$; it follows by applying concavity of the logarithm to $e^F(d\Pi/d\Gamma)$. Division by $a > 1/2$ proves (37). Infinite relative entropy makes the assertion vacuous and causes no exception. □

The target cylinder gives, for the actual posterior,

$$\frac{dQ_{S,N,t,c}}{d\mu_{N,t}} \le \exp(NK/t^2). \tag{38}$$

Here $K \ge 1$ may be the same sample-independent target-cost bound used by the reconstruction. In particular, $D_{\mathrm{KL}}(Q_{S,N,t,c}\|\mu_{N,t}) \le NK/t^2$. For the loss

$$\ell_w(x) = \mathbf{1}\{f(x)z_w(x) \le 1/4\},$$

the empirical loss is zero under this posterior. The following finite reference family makes the resulting risk bound simultaneous in width and scale; an infinite mixture over widths or exact scales is unnecessary.

Lemma 16 (A finite Gaussian reference family)**.** *Fix $K \ge 1$ and $J \ge K$. Set*

$$M = \lceil J/K \rceil, \qquad P_M = M(d+1) + (L-2)M^2 + M,$$

$$R_\tau = 1 + \left\lfloor \frac{P_M}{2} \log(J/K) \right\rfloor, \qquad \tau_j = \exp(-j/P_M) \quad (0 \le j < R_\tau).$$

There is a sample-independent prior on the disjoint union of these MR_τ Gaussian models such that, whenever $NK/t^2 \le J$, the original posterior, embedded in one component, has relative entropy at most

$$NK/t^2 + 1 + \log(MR_\tau). \tag{39}$$

Proof. The hypothesis implies $N \le M$ and $t^2 \ge K/J$. Choose $j = \lfloor P_M \log(1/t) \rfloor$. Then $0 \le j < R_\tau$, $\tau_j \ge t$, and $\log(\tau_j/t) \le 1/P_M$. At a fixed width all Gaussian covariance matrices are multiplied by the same scalar t^2. Their density ratio is therefore

$$\log \frac{d\mu_{N,t}}{d\mu_{N,\tau_j}}(w) = P_N \log(\tau_j/t) - \frac{N}{2}(t^{-2} - \tau_j^{-2})\kappa_N(w) \le P_N \log(\tau_j/t) \le 1.$$

The nonnegative weighted energy κ_N is the one defined in the main text. Combining this pointwise inequality with (38) bounds relative entropy with respect to μ_{N,τ_j} by $NK/t^2 + 1$. Choose the component (N, j) uniformly from the fixed finite family, then draw from its Gaussian law. Embedding the posterior in the chosen component adds exactly $\log(MR_\tau)$ and proves the claim. □

We next state the statistical interface needed from the reconstruction. Let ψ be the ramp loss used in the posterior comparison, and write

$$R_\psi(\Gamma) = \mathbb{E}_{h\sim\Gamma,X}\psi(f(X)T(h(X))), \qquad \widehat{R}_{\psi,S}(\Gamma) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}_{h\sim\Gamma}\psi(f(X_i)T(h(X_i))).$$

The readout-retention and pointwise coupling estimates give the two inequalities in the next theorem. Thus its hypotheses can be checked without imposing robustness on any decoded predictor.

Theorem 17 (A simultaneous width and scale bound)**.** *Suppose (38) holds. Fix nonnegative α, β and a message threshold Δ. Suppose there are sample-independent constants*

$$A \ge \max\{1, K, \Delta\}, \qquad B \ge 0,$$

with the following property. For every fixed $0 < a \le 1$ there is a finite class $\mathcal{H}_a$ satisfying

$$\log|\mathcal{H}_a| \le A + Ba^4. \tag{40}$$

Whenever $t \le a$ and $NK/t^2 > \Delta$, the reconstruction supplies a law Γ on $\mathcal{H}_a$ such that

$$\widehat{R}_{\psi,S}(\Gamma) \le \alpha, \qquad R(Q_{S,N,t,c}) \le \beta + \alpha + R_\psi(\Gamma). \tag{41}$$

Set $J = A + B$, use M, R_τ from Lemma 16, and define

$$j_a = \left\lceil \frac{1}{2}\log_2 \max\{1, B/A\} \right\rceil, \qquad R_a = j_a + 1, \qquad \Lambda = 1 + \log\frac{2MR_\tau R_a}{\delta}.$$

With sample probability at least $1 - \delta$, simultaneously for every $N \ge C$ and $0 < t \le 1$,

$$R(Q_{S,N,t,c}) \le \beta + 3\alpha + \frac{8}{n}\left[\min\{NK/t^2,\ A + Bt^4\} + \Lambda\right]. \tag{42}$$

On the same event, the branch $NK/t^2 \le \Delta$ obeys the sharper direct bound

$$R(Q_{S,N,t,c}) \le \frac{2}{n}[\Delta + \Lambda]. \tag{43}$$

Proof. Use the finite scale list $a_j = 2^{-j/2}$, $0 \le j < R_a$. For every $0 < t \le 1$, one of these upper scales satisfies $t \le a_j$ and

$$A + Ba_j^4 \le 4(A + Bt^4). \tag{44}$$

Indeed, above the smallest scale choose the nearest upper scale, whose square is less than twice t^2. Below the smallest scale, $Ba_{j_a}^4 \le A$ if $B > A$, while if $B \le A$ the single scale $a = 1$ already costs at most $2A$.

Construct one prior as follows. Give weight $1/2$ to the Gaussian reference family of Lemma 16. Give the remaining weight $1/2$ to a uniform choice of j, followed by the uniform law on $\mathcal{H}_{a_j}$. Keep the component tag as part of the hypothesis. For Gaussian components use the test-margin indicator loss; for finite code components use the ramp loss. These are a single $[0,1]$-valued loss on the tagged space. Apply Lemma 15 to this prior.

Put $b = NK/t^2$ and $F = A + Bt^4 \le J$. If $b \le F$, then $b \le J$, so the Gaussian family applies. The posterior has zero empirical indicator loss, and its relative-entropy charge, including the mixture weight and confidence term, is at most $b + \Lambda$. Thus its risk is at most $2(b + \Lambda)/n$.

If $b > F$, then $b > A \ge \Delta$, so the reconstruction applies with the scale selected in (44). Its relative-entropy charge is at most $4F + \Lambda$. Lemma 15 and (41) give risk at most $\beta + 3\alpha + 2(4F + \Lambda)/n$. These two cases imply (42). Finally, $b \le \Delta$ implies $b \le A \le J$, so the direct Gaussian calculation gives (43) on this same event. □

In particular, taking $\beta = \varepsilon/4$ and $\alpha = \varepsilon/12$, a sufficient sample condition at any given width and scale is

$$n \ge \frac{16}{\varepsilon}\left[\min\{NK/t^2,\ A + Bt^4\} + \Lambda\right]. \tag{45}$$

The event in Theorem 17 is common to all such pairs. For polynomially bounded A, B, K, d, L, the additional term Λ is logarithmic in these resources and in $1/\delta$.

Corollary 18 (Interpolation for stronger Gaussian scales)**.** *Under the hypotheses of Theorem 17, let $\gamma \ge 1$. On its same sample event, it suffices, simultaneously for every $N \ge C$ and $0 < t^2 \le N^{1-\gamma}$, to require*

$$n \ge \frac{16}{\varepsilon}\left[A + K^{(2\gamma-2)/(3\gamma-2)} B^{\gamma/(3\gamma-2)} + \Lambda\right]. \tag{46}$$

The convention for $B = 0$ is that its displayed contribution is zero.

Proof. If $B \le K$, then $A + Bt^4 \le A + B$ and $B \le K^{(2\gamma-2)/(3\gamma-2)} B^{\gamma/(3\gamma-2)}$; the result follows directly. Suppose $B > K$ and set

$$M_\gamma = (B/K)^{1/(3\gamma-2)}, \qquad u_0 = M_\gamma^{1-\gamma}, \qquad u = t^2.$$

If $u \le u_0$, the code term is at most $A + Bu_0^2$. If $u > u_0$ and $\gamma > 1$, the restriction $u \le N^{1-\gamma}$ implies $N < M_\gamma$, and therefore $NK/u \le KM_\gamma/u_0 = KM_\gamma^\gamma$. These two terms agree because

$$Bu_0^2 = KM_\gamma^\gamma = K^{(2\gamma-2)/(3\gamma-2)} B^{\gamma/(3\gamma-2)}.$$

For $\gamma = 1$, $u_0 = 1$ and only the first case is needed. Thus the minimum in (45) is at most the bracket in (46), proving the claim. □

Application to the entropy-metric code. The new message has log-cardinality

$$D_0 = [(d+1)q+2]\log 2 + q^2\log(3+8K_aq/e), \qquad K_a = 2K + \log(1/\eta), \quad q = \lceil K_a/e \rceil.$$

Take $\Delta = \max\{K, 2LD_0\}$. The large-branch condition in Theorem 17 is exactly the condition needed to absorb the accumulated messages under the conservative cutoff $N/t^2 > \max\{1, 2LD_0/K\}$. The complementary branch costs at most $\Delta = O(K + LD_0)$ by (43), instead of being rounded as a dense network of width LD_0/K.

For a fixed upper scale a, the variance padding can be chosen as $\lambda = \min\{1, e/(c_0a^2)\}$, where c_0 is a sufficiently large absolute constant. A separate uniform source sample of size $\widetilde{O}(1+\lambda^{-2}) = \widetilde{O}(1+a^4/e^2)$ estimates the total variance to accuracy λ. The extra physical Gaussian variance is $O(a^2\lambda) = O(e)$. Together with the active width $\widetilde{O}(K/e)$ and the quadratic message, the construction therefore gives constants in (40) with

$$A = \widetilde{O}(LK^2/e^2), \qquad B = \widetilde{O}(LK/e^3), \tag{47}$$

where A is enlarged to include Δ and the fixed lower-order terms. These are bounds on pre-sample constants, not on sample-dependent descriptions. In particular, the direct small branch is absorbed by the first term in (47).

For fixed d, c, ε, put $F_0 = 1 + \log^2(2+C)$ and $G = \log(2+CL)$. The target-cylinder estimates give $K = O(CLF_0)$ and $e^{-1} = O(CL^3F_0)$. Consequently $q = O(C^2L^4F_0^2)$, $D_0 = O(q^2G)$, and the total source width at upper scale a is

$$m = O\left([C^2L^4F_0^2 + a^4C^2L^6F_0^2]G\right).$$

The scalar count is $O(L[mq+q^2])$, up to lower-order first-layer terms. Polynomial coefficient bounds and inverse meshes contribute one further factor G, uniformly in $a \le 1$: in particular $\lambda^{-1} \le c_0/e$. Thus, after increasing a constant depending only on d, c, ε, the choices in (47) satisfy

$$A = O(C^4L^9F_0^4G^2), \qquad B = O(C^4L^{11}F_0^4G^2). \tag{48}$$

Moreover $\Lambda = O(G + \log(1/\delta))$. The width and scale envelope becomes

$$\widetilde{O}\min\{CLN/t^2,\ C^4L^9 + t^4C^4L^{11}\}.$$

Corollary 18 gives the sufficient sample order

$$\widetilde{O}\left(C^4L^9 + C^{(6\gamma-2)/(3\gamma-2)}L^{(13\gamma-2)/(3\gamma-2)}\right). \tag{49}$$

An explicit common logarithmic factor in (49) is $F_0^4G^2$, with an additional additive $O(\log(1/\delta))$ term before the accuracy-dependent constant. Indeed, the interpolated term has F_0 exponent $(6\gamma-2)/(3\gamma-2) \le 4$ and G exponent $2\gamma/(3\gamma-2) \le 2$ for every $\gamma \ge 1$. At $\gamma = 1$ this is $\widetilde{O}(C^4L^{11})$. For every $\gamma \ge 1$ the second term's width exponent is at most four; its depth exponent is at most nine exactly when $\gamma \ge 8/7$. Thus $\widetilde{O}(C^4L^9)$ suffices throughout the regime $0 < t^2 \le N^{-1/7}$. The existing stronger-scale theorem remains sharper at $0 < t^2 \le N^{-1}$, where it gives $\widetilde{O}(C^4L^7)$.

The direct posterior argument uses only the target-cylinder denominator and zero empirical margin loss. It therefore also applies to the optional global clean-margin conditioning whenever the same cylinder is retained.

F A multilevel code for two-step reconstructions

This appendix gives another finite class of representatives for the original robustness support. Its members are finite programs using arithmetic, tanh, Euclidean projection, and a fixed finite median rule. They need not themselves be robust networks. All randomness below is used only to prove the existence of a representative in a fixed finite class; the representative is deterministic after its coefficients and finite computation graph have been recorded.

Theorem 19 (Multilevel finite code)**.** *There is a universal constant A with the following property. Let $d, H \geq 1$, $K > 0$, and $0 < r \leq 1/6$, and put*

$$E = K + H + 1, \qquad T_0 = d + \log(2H + 2), \qquad \Lambda = \log(2 + d + E + r^{-1}).$$

There is a finite class $\mathcal{G}_{K,r,d,H}$, independent of learner width, such that

$$\log |\mathcal{G}_{K,r,d,H}| \leq AT_0 \left[\frac{E^3}{r^3} + \frac{(d+1)E^2}{r^2} \right] \Lambda. \tag{50}$$

For every width N and every network w with $\kappa_N(w) \leq K$, there is $g \in \mathcal{G}_{K,r,d,H}$ whose output is the output of the original network with input-dependent disturbances of stacked norm at most r. Consequently, if $w \in \mathcal{R}_{N,c}$ and $r \leq c/\sqrt{1 + HE_v(w)}$, then

$$\sup_{x \in \{-1,1\}^d} |T(z_w(x)) - T(g(x))| \leq 1/16.$$

We prove the theorem in three steps. First we construct an economical code for a map containing two activations. We then embed such maps in a virtual computation of the original network. Finally we choose spectral cutoffs and error allocations jointly across layers.

F.1 A two-activation sampling lemma

Consider the vector-valued map

$$F(z, z') = \sum_{i=1}^{N} w_i \tanh \left(a_i \cdot z + \sum_{j=1}^{N} R_{ij} \tanh(c_j \cdot z') \right), \qquad w_i \in \mathbb{R}^{q_o}, \tag{51}$$

where $a_i \in \mathbb{R}^{q_a}$ and $c_j \in \mathbb{R}^{q_c}$. Assume

$$N \sum_i \|w_i\|_2^2 \leq k_o, \quad \|R\|_F^2 \leq k_R, \quad \frac{1}{N} \sum_i \|a_i\|_2^2 \leq k_a, \quad \frac{1}{N} \sum_j \|c_j\|_2^2 \leq k_c. \tag{52}$$

Write

$$A_i = \sum_j |R_{ij}|, \qquad M_i = \max\{1, A_i\}, \qquad B = \sum_i \|w_i\|_2 ,$$

$$D = \sum_i \|w_i\|_2 M_i, \qquad Z = \sum_i \frac{\|w_i\|_2}{M_i}. \tag{53}$$

Cauchy–Schwarz gives

$$B \leq \sqrt{k_o}, \qquad B^2 \leq ZD, \qquad D^2 \leq 2k_o(1 + k_R). \tag{54}$$

Indeed, $\sum_i A_i^2 \leq N \|R\|_F^2$, so $D \leq \sqrt{k_o}(1 + \sqrt{k_R})$.

Lemma 20 (Antithetic map sampling)**.** *Fix $\tau > 0$ and a finite collection of evaluation arguments. There is a random finite program approximating (51) with the following properties. At each fixed argument, its error is at most 9τ except on an event of probability at most $e^{-c_1 g}$, where g is the odd number of independent median groups. Its expected number of stored scalar coefficients is at most*

$$c_2 g(1 + q_a + q_c + q_o)\frac{k_o(1 + k_R)}{\tau^2}. \tag{55}$$

After an event of constant probability bounding total coefficient work, the program can be rounded to a fixed grid with one further error at most a constant times τ. All parameter ranges and the inverse grid mesh are polynomial in the energy and dimension upper bounds, the evaluation-argument norm bounds, τ^{-1}, and the work bound. No original-width neuron index is needed in the stored program.

The probability assertion also holds conditionally when the evaluation arguments have been produced by earlier independent random programs.

Proof. We give the sampling, coefficient, and rounding details. Deterministic feature pruning will first ensure finite coefficient ranges. Removing roots with $\|a_i\|_2 > B_a$ changes the map by at most

$$\sum_{\|a_i\|_2 > B_a} \|w_i\|_2 \le \sqrt{k_o k_a}/B_a.$$

Deleting columns j with $\|c_j\|_2 > B_c$ from R changes it by at most

$$\begin{aligned}\sum_i \|w_i\|_2 \sum_{j:\|c_j\|_2 > B_c} |R_{ij}| &\le \sqrt{k_o/N}\,\sqrt{\#\{j : \|c_j\|_2 > B_c\}\, k_R} \\ &\le \sqrt{k_o k_R k_c}/B_c.\end{aligned}$$

Choose the two cutoffs to make these errors at most τ each; zero numerators require no cutoff. All old energy upper bounds remain valid after pruning. Recompute A_i, M_i, B, D, Z for the pruned map. If $B \le \tau$, the zero map already has the required accuracy, so assume $B > \tau$.

For $A_i > 0$, sample J with probability $|R_{iJ}|/A_i$ and put

$$X_i = A_i \operatorname{sgn}(R_{iJ}) \tanh(c_J \cdot z').$$

Then $|X_i| \le A_i$ and $\mathbb{E}X_i = \mu_i := \sum_j R_{ij} \tanh(c_j \cdot z')$. For $A_i = 0$, use $X_i = 0$. Let

$$n_{i,0} = \lceil M_i^2 \rceil, \qquad n_{i,l} = 2^l n_{i,0}.$$

At level zero define $\Delta_{i,0} = \tanh(a_i \cdot z + \overline{X}_{i,n_{i,0}})$. At level $l \ge 1$, split $n_{i,l}$ independent observations into two equal halves with averages Y, Y', and define

$$\Delta_{i,l} = \tanh\left(a_i \cdot z + \frac{Y + Y'}{2}\right) - \frac{1}{2}\tanh(a_i \cdot z + Y) - \frac{1}{2}\tanh(a_i \cdot z + Y'). \tag{56}$$

All three evaluations reuse the same child activations. The bounded second derivative of tanh and the fourth moment of a bounded sample average imply, with universal constants,

$$\mathbb{E}|\Delta_{i,l}|^2 \le c_3 2^{-2l}, \qquad \left|\mathbb{E}\tanh(a_i \cdot z + \overline{X}_{i,n_{i,l}}) - \tanh(a_i \cdot z + \mu_i)\right| \le c_4 2^{-l}. \tag{57}$$

For completeness, Taylor's theorem bounds the absolute value in (56) by $\|\tanh''\|_\infty |Y - Y'|^2/8$. The fourth moment of a centered average of m independent variables bounded by $2A_i$ is at most a

universal constant times A_i^4/m^2. Since $n_{i,0} \geq A_i^2$, this proves the first estimate. The second follows from the same Taylor bound and the variance of the average. The estimates are uniform in both evaluation arguments.

Choose a finite last level l_* and let

$$p_i = \frac{\|w_i\|_2}{M_i Z}, \qquad \pi_l = \frac{2^{-3l/2}}{\sum_{j=0}^{l_*} 2^{-3j/2}}.$$

Zero-weight roots are omitted. Independently choose a root and level, and return the vector $Y = w_i \Delta_{i,l}/(p_i \pi_l)$. The antithetic expectations telescope. Therefore, if $F_* = \mathbb{E}Y$,

$$\|F_* - F\|_2 \leq c_4 B 2^{-l_*}, \qquad \mathbb{E}\,\|Y\|_2^2 \leq c_5 ZD, \qquad \mathbb{E}\,\mathrm{work}(Y) \leq c_6 D/Z. \tag{58}$$

Here scalar work counts child activations, with a fixed constant for each root. The identities proving the last two estimates are

$$\sum_i \frac{\|w_i\|_2^2}{p_i} = ZD, \quad \sum_i p_i M_i^2 = D/Z, \quad \sum_l 2^{-2l}/\pi_l = O(1), \quad \sum_l \pi_l 2^l = O(1).$$

Choose $2^{l_*} \geq c_4\sqrt{k_o}/\tau$, or a larger common polynomial upper bound, making the bias at most τ.

Average $m = \lceil c_7 ZD/\tau^2 \rceil$ independent copies of Y in each group. For sufficiently large c_7, Chebyshev's inequality makes the probability of vector error exceeding τ about F_* at most $1/64$. The expected work of a group is

$$O(D^2/\tau^2 + D/Z) = O(D^2/\tau^2),$$

because $B > \tau$ and $B^2 \leq ZD$ imply $D/Z \leq D^2/B^2 < D^2/\tau^2$.

Use g independent groups and the following fixed median rule: choose the first group output with at least $(g+1)/2$ group outputs within distance 4τ, and return zero if there is no such group. Except with probability $e^{-c_1 g}$, a strict majority of the groups lie within τ of F_*. Every such group qualifies, and every selected group lies within 5τ of F_*. If each group output is subsequently changed by at most τ, the same rule returns a point within 6τ of F_*. This follows because the qualifying ball must intersect the strict majority of good groups. Thus no Lipschitz property of the median operation is needed. Combining the two pruning errors, finite-level bias, and the rounded median gives error at most 9τ.

Multiplying scalar work by $1+q_a+q_c+q_o$ bounds the number of recorded scalar coefficients: record each selected root feature, each child feature, its signed inner coefficient, and the vector coefficient of each term in a group average. This proves (55). For several maps, apply Markov's inequality to their *total dimension-weighted coefficient work*, rather than separately to their unweighted child counts. On an event of constant probability, this total is at most its chosen common deterministic bound W. Every selected root then has $M_i^2 \leq n_{i,l} \leq W$.

The vector coefficient of an averaged term is bounded by

$$\left\| \frac{w_i}{m p_i \pi_l} \right\|_2 \leq \frac{\tau^2 M_i}{c_7 D \pi_l} \leq \frac{\tau\sqrt{W}}{c_7 \pi_l},$$

using $D \geq B > \tau$. Inner coefficients have magnitude at most $A_i \leq \sqrt{W}$. Feature pruning bounded the remaining feature coefficients, and the finite-level inverse probabilities are polynomial in the common resources. Thus every recorded coefficient lies in a fixed polynomially bounded interval.

Suppose all evaluation arguments have norm at most a fixed polynomial bound. There are at most W stored coefficients and only two nested activation levels. The 1-Lipschitz property of tanh shows that rounding all coefficients to a sufficiently fine reciprocal-polynomial mesh changes every group output by at most τ, uniformly on these arguments. The preceding median argument applies after this rounding. The forest topology, its group assignment, and its half-list divisions have description length $O(W \log(2+W))$. Only numeric feature vectors are stored, so no original-width neuron label appears. Finally, all stochastic estimates were uniform at a fixed argument. Conditioning on arguments produced by independent preceding programs therefore preserves them, and a union bound applies to any finite collection of arguments. □

F.2 Realization inside the original robustness ball

Write $|h|_N = \|h\|_2 / \sqrt{N}$ and let $k_s = \|W^s\|_F^2$ for $2 \le s \le H$, with $k_{H+1} = E_v(w)$. Split each internal matrix by singular value truncation at a threshold $a_s \in (0,1)$:

$$W^s = A_s + R_s, \qquad A_s = U_s \Sigma_s V_s^\top, \qquad \|R_s\|_{\mathrm{op}} \le a_s, \qquad q_s = \operatorname{rank} A_s \le k_s / a_s^2.$$

For drive coordinates $t_s \in \mathbb{R}^{q_s}$ use the original-space vector $u_s = \sqrt{N} U_s t_s$. An exact low-rank drive of a tanh state h is $t_s = \Sigma_s V_s^\top h / \sqrt{N}$, of norm at most $\sqrt{k_s}$.

The first drive is the one-activation map $\Sigma_2 V_2^\top h^1(x)/\sqrt{N}$. The next drive uses $h^1(x)$ as its left boundary. All later drive maps have the two-activation form

$$\frac{1}{\sqrt{N}} \Sigma_s V_s^\top \tanh\left(\sqrt{N} U_{s-1} t_{s-1} + R_{s-1} \tanh(\sqrt{N} U_{s-2} t_{s-2})\right). \tag{59}$$

In Lemma 20, the output-energy bound is k_s, the residual-energy bound is k_{s-1}, and the root and child feature second moments are q_{s-1} and q_{s-2}. For an early map using the original first state, its child features are rows of W^1, whose normalized squared norm sum is at most K/d. The final readout has the same form with scalar output and output-energy bound k_{H+1}.

Construct and round each map while advancing the computation. After each drive map, project its output onto the ball of radius $\sqrt{\max(K, H, 1)}$. This does not increase error about its exact target drive. Let $\widehat{t}_s$ be the resulting coordinates and define full virtual states by

$$\widehat{h}^1 = h^1, \qquad \widehat{h}^2 = \tanh(\sqrt{N} U_2 \widehat{t}_2 + R_2 h^1),$$
$$\widehat{h}^s = \tanh\left(\sqrt{N} U_s \widehat{t}_s + R_s \tanh(\sqrt{N} U_{s-1} \widehat{t}_{s-1})\right), \qquad s \ge 3. \tag{60}$$

These full states are used only to verify the disturbance bound; the stored program advances the drive coordinates through its finite map codes and evaluates the final readout code.

Suppose the drive-map errors are at most δ_s, and the final readout-map error is at most δ_{H+1}. Their exact targets are the low-rank drives of the already rounded preceding virtual states. For $s \ge 3$, comparison with the original recurrence gives

$$|\widehat{h}^s - \tanh(W^s \widehat{h}^{s-1})|_N \le \delta_s + a_s a_{s-1}.$$

Indeed, the low-rank preactivation discrepancy has normalized norm at most δ_s, while the discarded older residual passes through R_{s-1} and R_s, with a 1-Lipschitz tanh between. Its left boundary is a tanh state of normalized norm at most one. At stage two the bound is just δ_2; the first-state disturbance is zero. Consequently the original-network disturbances realizing the virtual output satisfy

$$\|\eta\|_* \le \left(\sum_{s=2}^{H+1} \delta_s^2\right)^{1/2} + \left(\sum_{s=3}^{H} a_s^2 a_{s-1}^2\right)^{1/2}. \tag{61}$$

For $H = 1$, only the one-activation readout map is needed; for $H = 2$, the exact first-state boundary is retained throughout.

Every stage uses fresh map samples independent of earlier map programs. Lemma 20 therefore applies conditionally at the already rounded prefix. Choose $g = O(T_0)$ so a union bound over all 2^d inputs and all maps fails with probability at most $1/100$. The coefficient-work event can be chosen to fail with probability at most $1/100$, so the two events intersect. All rounding is already included in the map errors in (61).

F.3 Adaptive cutoffs and a common finite class

Put $e_s = 1 + k_s$ for the internal blocks and the readout. Their sum is at most $E = K + H + 1$. For a sufficiently small universal constant c_0, choose

$$a_s^2 = \frac{c_0 r \sqrt{e_s}}{\sqrt{E}}, \qquad 2 \le s \le H. \tag{62}$$

Then $a_s < 1$ and

$$\sum_{s=3}^{H} a_s^2 a_{s-1}^2 = \frac{c_0^2 r^2}{E} \sum_{s=3}^{H} \sqrt{e_s e_{s-1}} \le c_0^2 r^2.$$

With $\alpha = \sqrt{E}/(c_0 r)$, the retained ranks satisfy $q_s \le \alpha \sqrt{e_s}$.

Ignoring fixed constants, define each map's coefficient resource as

$$B_s = (1 + q_{s-2} + q_{s-1} + q_s) k_s (1 + k_{s-1}). \tag{63}$$

At the first two maps, replace the missing input dimension by $d + 1$ and omit a nonexistent residual factor. At the readout, replace the output dimension by one and the output energy by k_{H+1}. These conventions give exactly the resources in Lemma 20. Apart from the first two original- input contributions, they are bounded by a universal constant times

$$\alpha e_s e_{s-1} (\sqrt{e_{s-2}} + \sqrt{e_{s-1}} + \sqrt{e_s}).$$

At boundary indices in this estimate, set a missing e_j equal to one; the finitely many auxiliary indices change the following bounds only by universal constants. For each of the three indicated shifts j,

$$\sum_s e_s^{1/2} e_{s-1}^{1/2} e_j^{1/4} \le (\max_j e_j)^{1/4} \sum_s \sqrt{e_s e_{s-1}} \le c_8' E^{5/4}.$$

Using $\sqrt{u + v + w} \le \sqrt{u} + \sqrt{v} + \sqrt{w}$, and treating the two original-input terms separately, gives

$$\left(\sum_s \sqrt{\max(1, B_s)} \right)^2 \le c_8 \left[\frac{E^3}{r} + (d + 1) E^2 \right]. \tag{64}$$

Inserting the ones increases the unsquared sum by at most $H + 1$, so its additional squared contribution is $O(H^2)$, already absorbed by the right side. Each original-input term contributes at most a constant times $\sqrt{d + 1}\, E$ before squaring.

Let S_0 be the square root of the right side of (64), enlarged by a fixed constant. Set

$$\delta_s^2 = \frac{r^2}{64 S_0} \sqrt{\max(1, B_s)},$$

and round each δ_s downward to a power-of-two multiple of a fixed reciprocal-polynomial base, losing at most a factor two. Thus $\sum_s \delta_s^2 \le r^2/64$, every nonzero tolerance has a common reciprocal-polynomial lower bound, and

$$\sum_s B_s \delta_s^{-2} \le c_9 \left[\frac{E^3}{r^3} + \frac{(d+1)E^2}{r^2} \right]. \tag{65}$$

Apply the sampling lemma with τ_s a sufficiently small fixed multiple of δ_s. Equations (61) and (62) put the complete virtual computation inside the radius-r disturbance ball.

To make the code common to all widths and all retained networks, record the integer rank profile, the dyadic tolerance choices, and the finite map forests. Every rank is at most $E/(c_0 r)$. The bounds (65) and $g = O(T_0)$ give a common polynomial bound W on total coefficient work on the chosen Markov event. Feature cutoffs, coefficient magnitudes, inverse grid mesh, and finite last levels are all polynomial in $2 + d + E + r^{-1}$, with universal degrees: this follows from the common rank bound, the common lower tolerance bound, and the coefficient estimates in the sampling lemma. Fix those ranges and that grid before defining the class of all recorded programs. The possible topology, ranks, and tolerance choices cost $O(W\Lambda + H\Lambda)$ nats; the coefficient grid costs $O(W\Lambda)$. Since $H \le E$ and $r \le 1/6$, the former additional term is absorbed. This proves (50). The class contains no learner-width labels, so the same class works for every N. This completes the proof of Theorem 19.

F.4 The original posterior at the stronger scales

The new code changes neither the prior nor the conditioning event. The following statement records the exact scale hypotheses used with it. Use the product-cylinder cost $D_* = D_*^{\mathrm{prod}}$ from (17), and write

$$P_N = N(d+1) + (L-2)N^2 + N.$$

For the chosen energy cutoff K, retain the radius

$$r = \min\{1/6, c/\sqrt{1+HK}\}.$$

Let J_{ML} be the upper bound in (50), and let J be the earlier dense-code budget in (4), with this same K and r. Put

$$J_{\mathrm{best}} = \min\{J, J_{\mathrm{ML}}\}.$$

This minimum is legitimate without a union penalty: choose, from the known resource bounds, whichever of the two common finite classes has the smaller stated cardinality bound.

Theorem 21 (A mixed width–depth bound at the original stronger scales)**.** *Fix $0 < \varepsilon, \delta < 1$. Suppose the target has the Boolean implementation specified in the paper. The sample condition*

$$n \ge \frac{2}{\varepsilon}\left[J_{\mathrm{best}} + \log(1/\delta)\right]$$

implies, with sample probability at least $1-\delta$, $R(Q_{S,N,t,c}) \le \varepsilon$ simultaneously at every indicated width and scale in either of the following regimes:

1. *Set $K = 8D_*$. Allow every $N \ge C$ and every positive scale with*

$$t^2 \le \min\left\{1, \frac{ND_*}{(P_N/2)\log 2 + \log(2/\varepsilon)}\right\}.$$

2. Set

$$K = 8D_* + 4(d+L) + 4\log(2/\varepsilon).$$

Allow every $N \geq C$ and $0 < t \leq N^{-1/2}$, in particular every power-law scale with $\gamma \geq 2$.

In the first regime, at each fixed width the same sample event gives $\limsup_{t\downarrow 0} R(Q_{S,N,t,c}) \leq \varepsilon$. At fixed $d, c, \varepsilon, \delta$, a sufficient sample count is

$$\widetilde{O}\Big(\min\{C^4L^7, C^{9/2}L^6\}\Big).$$

Proof. Every robust network with $\kappa_N(w) \leq K$ has calibrated radius at least r. Theorem 19, or the earlier dense code, therefore supplies a representative with clipped-score error at most $1/16$ at every Boolean input. The representative need not be a robust network; only its membership in a common finite class is used next.

Use the ramp loss that is one at signed clipped scores at most $1/3$ and zero at scores at least $5/12$. Every representative of a fitting network has zero empirical ramp loss, because its signed clipped training score is at least $7/16$. At every test-margin failure of the original network, its representative has signed clipped score at most $5/16$, so the ramp loss is one. A union bound over the chosen finite class shows that, on a single sample event of probability at least $1-\delta$, every retained fitting network has test-margin error at most $\varepsilon/2$. This sample event is independent of learner width and Gaussian scale.

The product-cylinder lower bound and the Gaussian exponential moment give the same tail as (11), with the current valid cylinder cost:

$$Q_{S,N,t,c}\{\kappa_N > K\} \leq \exp\Big[-\frac{NK}{4t^2} + \frac{P_N}{2}\log 2 + \frac{ND_*}{t^2}\Big].$$

In the first regime the displayed scale inequality makes this at most $\varepsilon/2$. In the second regime use $t^2P_N/N \leq d+L$ and the stated value of K, exactly as in the proof of Theorem 3. Adding the discarded mass proves the risk claim. At fixed N, the first scale inequality holds for all sufficiently small positive t, proving the stated strong-prior conclusion without identifying a separately defined zero-temperature posterior.

Finally, the product-cylinder calculation gives $K = \widetilde{O}(CL)$ and $r^{-2} = \widetilde{O}(CL^2)$. The budget (50), with $K + H + 1 = \widetilde{O}(CL)$, is therefore $\widetilde{O}(C^{9/2}L^6)$ at the fixed resources stated in the theorem. Taking the better of this and the existing $\widetilde{O}(C^4L^7)$ code gives the final display. □

Corollary 22 (Explicit logarithms in the mixed-power budget)**.** *Fix $d, c, \varepsilon, \delta$, and put $F_0 = 1 + \log^2(2+C)$ and $G = \log(2+CL)$. In either regime of Theorem 21, a sufficient sample count is*

$$O\Big(G\min\Big\{C^4L^7F_0^4\log^2(2+L),\ C^{9/2}L^6F_0^{9/2}\log(2+L)\Big\}\Big).$$

In particular, at fixed teacher width the new branch is $O(L^6\log^2(2+L))$.

Proof. The product-cylinder cost satisfies $D_*^{\mathrm{prod}} = O(CLF_0)$. For either stated energy cutoff,

$$E = O(CLF_0), \qquad r^{-2} = O(CL^2F_0), \qquad T_0 = O(\log(2+L)), \qquad \Lambda = O(G).$$

Consequently the leading term in (50) is

$$O\Big(C^{9/2}L^6F_0^{9/2}\log(2+L)G\Big).$$

The term $(d+1)E^2r^{-2}$ has order $O(C^3L^4F_0^3)$ and is smaller at the fixed ancillary resources. The dense-code calculation with the same product-cylinder substitution gives $O(C^4L^7F_0^4\log^2(2+L)G)$, as recorded after Corollary 6. Taking the better code and applying Theorem 21 proves the result. □

The second branch improves the stated depth dependence at fixed teacher width, and is smaller at the level of the displayed powers when $C < L^2$. It does not establish a better uniform quartic-width exponent. No bounded spectral product, column coherence, additional output-margin condition, or modified robustness support is used.

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